My Humble Solution to

Volume 1 of Course of Theoretical Physics

Mechanics

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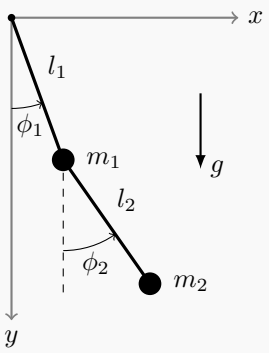
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Chapter 1.

The Equations of Motion

Problem 1.

Find the Lagrangian of a coplanar double pendulum when placed in a uniform gravitational field (acceleration g).



Solution: The generalized co-ordinates of the system are the two angles ϕ_1 and ϕ_2. We need to express the Cartesian co-ordinates in terms of those two angles. First, the Cartesian position of the particle m_1 is

$$\begin{aligned} x_1 &= f_1(\phi_1) = l_1 \sin\phi_1 \\ y_1 &= g_1(\phi_1) = l_1 \cos\phi_1. \end{aligned}$$

By taking the time derivative of those, we obtain

$$\begin{aligned} \dot{x}_1 &= \frac{\partial f_1}{\partial \phi_1}\dot{\phi}_1 = l_1 \cos\phi_1 \dot{\phi}_1 \\ \dot{y}_1 &= \frac{\partial g_1}{\partial \phi_1}\dot{\phi}_1 = -l_1 \sin\phi_1 \dot{\phi}_1. \end{aligned}$$

Substituting these expressions in the Lagrangian, we obtain the desired Lagrangian form, that is

$$\begin{aligned} L_1 &= T_1 - U_1 \\ &= \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + m_1 g y_1 \\ &= \frac{1}{2} m_1 l_1^2 \dot{\phi}_1^2 + m_1 g l_1 \cos\phi_1. \end{aligned} \tag{1.1}$$

Second, the Cartesian position of the particle m_2 is

$$\begin{aligned} x_2 &= f_2(\phi_1, \phi_2) = l_1 \sin\phi_1 + l_2 \sin\phi_2 \\ y_2 &= g_2(\phi_1, \phi_2) = l_1 \cos\phi_1 + l_2 \cos\phi_2. \end{aligned}$$

By taking the time derivative of those, we obtain

$$\begin{aligned} \dot{x}_2 &= \sum_k \frac{\partial f_2}{\partial \phi_k}\dot{\phi}_k = l_1 \dot{\phi}_1 \cos\phi_1 + l_2 \dot{\phi}_2 \cos\phi_2 \\ \dot{y}_2 &= \sum_k \frac{\partial g_2}{\partial \phi_k}\dot{\phi}_k = -l_1 \dot{\phi}_1 \sin\phi_1 - l_2 \dot{\phi}_2 \sin\phi_2. \end{aligned}$$

Substituting these expressions in the Lagrangian, we obtain the desired Lagrangian form, that is

$$\begin{aligned} L_2 &= T_2 - U_2 \\ &= \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2) + m_2 g y_2 \\ &= \frac{1}{2} m_2 (l_1^2 \dot{\phi}_1^2 + l_2^2 \dot{\phi}_2^2 + 2 l_1 l_2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2)) + m_2 g (l_1 \cos\phi_1 + l_2 \cos\phi_2), \end{aligned} \tag{1.2}$$

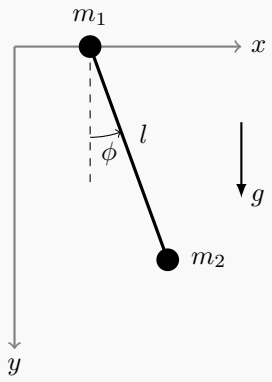
where the angle difference identity as been used. Finally, the Lagrangian of the complete system is simply the sum of Eq. (1.1) and Eq. (1.2), thus

Answer to Problem 1.

$$\begin{aligned} L =& \frac{1}{2}(m_1+m_2)l_1^2\dot{\phi}_1^2 + \frac{1}{2}m_2 l_2^2\dot{\phi}_2^2 + m_2 l_1 l_2 \dot{\phi}_1\dot{\phi}_2\cos(\phi_1-\phi_2) \\ &+ (m_1+m_2)gl_1\cos\phi_1 + m_2 g l_2\cos\phi_2 \end{aligned}$$

Problem 2.

Find the Lagrangian of a simple pendulum of mass m_2, with a mass m_1 at the point of support which can move on a horizontal line lying in the plane in which m_2 moves when placed in a uniform gravitational field (acceleration g).



Solution: The generalized co-ordinates q of the system are the position x and the angle ϕ. Therefore, the Lagrangian of the first particle is simply

$$L_1 = T_1 = \frac{1}{2} m_1 \dot{x}^2. \tag{1.3}$$

The Cartesian position of the particle m_2 can be express in terms of the generalized co-ordinates by

$$\begin{aligned} x_2 &= f_2(x, \phi) = x + l \sin \phi \\ y_2 &= g_2(x, \phi) = l \cos \phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x}_2 &= \sum_k \frac{\partial f_2}{\partial q_k} \dot{q}_k = \dot{x} + l\dot{\phi} \cos \phi \\ \dot{y}_2 &= \sum_k \frac{\partial g_2}{\partial q_k} \dot{q}_k = -l\dot{\phi} \sin \phi. \end{aligned}$$

$$\begin{aligned} L_2 &= T_2 - U_2 \\ &= \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2) + m_2 g y_2 \\ &= \frac{1}{2} m_2 (\dot{x}^2 + l^2 \dot{\phi}^2 + 2l\dot{x}\dot{\phi} \cos \phi) + m_2 g l \cos \phi. \end{aligned} \tag{1.4}$$

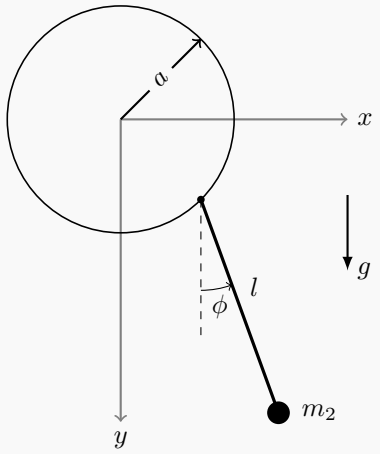
Finally, the Lagrangian of the complete system is the sum of the two Lagrangian Eq. (1.3) and Eq. (1.4), thus

Answer to Problem 2.

$$L = \frac{1}{2}(m_1 + m_2)\dot{x}^2 + \frac{1}{2} m_2 (l^2 \dot{\phi}^2 + 2l\dot{x}\dot{\phi} \cos \phi) + m_2 g l \cos \phi$$

Problem 3.

Find the Lagrangian of a simple pendulum of mass m, when placed in a uniform gravitational field (acceleration g), whose point of support ...



(a)

moves uniformly on a vertical circle with constant frequency γ.

Solution: If we set that the rotation of the point of support is counterclockwise, the Cartesian position of the particle m is

$$\begin{aligned} x &= f(\phi_1) = a\cos\gamma t + l\sin\phi \\ y &= g(\phi_1) = -a\sin\gamma t + l\cos\phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x} &= -a\gamma\sin\gamma t + l\dot{\phi}\cos\phi \\ \dot{y} &= -a\gamma\cos\gamma t - l\dot{\phi}\sin\phi. \end{aligned}$$

The potential energy of the system is

$$\begin{aligned} U &= -mgy \\ &= -mg(-a\sin\gamma t + l\cos\phi). \end{aligned}$$

The term $mga\sin\gamma t$ only depend on time and can therefore be ignored (does not contribute to the equations of motion). The kinetic energy of the particle, for its part, is

$$\begin{aligned} T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \frac{1}{2}m\left(l^2\dot{\phi}^2 + a^2\gamma^2 + 2a\gamma l\dot{\phi}\sin\left(\phi - \gamma t\right)\right), \end{aligned}$$

using again the angle difference identity (I will stop mentioning it). We can observe that the term $\frac{1}{2}ma^2\gamma^2$ is a constant, thus can be ignored. The last term of the kinetic energy can also be simplified. Indeed,

$$\begin{aligned} ma\gamma l\dot{\phi}\sin\left(\phi - \gamma t\right) &= mal\gamma(\dot{\phi} - \gamma)\sin(\phi - \gamma t) + mal\gamma^2\sin(\gamma t - \phi) \\ &= \frac{d}{dt}\left(-mal\gamma\cos(\phi - \gamma t)\right) + mal\gamma^2\sin(\phi - \gamma t). \end{aligned}$$

After dropping the total time derivative, we can finally get the Lagrangian

Answer to Problem 3 (a).

$$L = \frac{1}{2}ml^2\dot{\phi}^2 + mal\gamma^2\sin(\phi - \gamma t) + mgl\cos\phi$$

(b)

oscillates horizontally in the plane of motion of the pendulum according to the law $x = a\cos\gamma t$.

Solution: The Cartesian position of the particle m is

$$\begin{aligned} x &= f(\phi_1) = a\cos\gamma t + l\sin\phi \\ y &= g(\phi_1) = l\cos\phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x} &= -a\gamma\sin\gamma t + l\dot{\phi}\cos\phi \\ \dot{y} &= -l\dot{\phi}\sin\phi. \end{aligned}$$

The potential energy of the system is

$$U = -mgy = -mgl\cos\phi,$$

and the kinetic energy

$$\begin{aligned} T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \frac{1}{2}m\left(l^2\dot{\phi}^2 + a^2\gamma^2\sin^2\gamma t - 2a\gamma l\dot{\phi}\sin\gamma t\cos\phi\right). \end{aligned}$$

We first see that the term $\frac{1}{2}ma^2\gamma^2\sin^2\gamma t$ only depend on time. The last term of the kinetic energy can also be simplified. Indeed,

$$\begin{aligned} -ma\gamma l\dot{\phi}\sin\gamma t\cos\phi &= -mla\gamma(\gamma\cos\gamma t\sin\phi + \dot{\phi}\sin\gamma t\cos\phi) + mla\gamma^2\cos\gamma t\sin\phi \\ &= \frac{d}{dt}\left(-mla\gamma\sin\gamma t\sin\phi\right) + mla\gamma^2\cos\gamma t\sin\phi. \end{aligned}$$

After dropping the total time derivative, we can finally get the Lagrangian

Answer to Problem 3 (b).

$$L = \frac{1}{2}ml^2\dot{\phi}^2 + mla\gamma^2\cos\gamma t\sin\phi + mgl\cos\phi$$

(c)

oscillates vertically according to the law $y = a\cos\gamma t$.

Solution: The Cartesian position of the particle m is

$$\begin{aligned} x &= f(\phi_1) = l\sin\phi \\ y &= g(\phi_1) = a\cos\gamma t + l\cos\phi. \end{aligned}$$

The time derivative of the position is then

$$\begin{aligned} \dot{x} &= l\dot{\phi}\cos\phi \\ \dot{y} &= -a\gamma\sin\gamma t - l\dot{\phi}\sin\phi. \end{aligned}$$

The potential energy of the system is

$$U = -mgy = -mg\left(a\cos\gamma t + l\cos\phi\right),$$

The term $mga\cos\gamma t$ only depend on time and can therefore be ignored. The kinetic energy is

$$\begin{aligned} T &= \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) \\ &= \frac{1}{2}m\left(l^2\dot{\phi}^2 + a^2\gamma^2\sin^2\gamma t + 2a\gamma l\dot{\phi}\sin\gamma t\sin\phi\right). \end{aligned}$$

We first see that the term $\frac{1}{2}ma^2\gamma^2\sin^2\gamma t$ only depend on time. The last term of the kinetic energy can also be simplified. Indeed,

$$\begin{aligned} ma\gamma l\dot{\phi}\sin\gamma t\sin\phi &= mla\gamma(-\gamma\cos\gamma t\cos\phi + \dot{\phi}\sin\gamma t\sin\phi) + mla\gamma^2\cos\gamma t\cos\phi \\ &= \frac{d}{dt}\left(mla\gamma\cos\gamma t\sin\phi\right) + mla\gamma^2\cos\gamma t\cos\phi. \end{aligned}$$

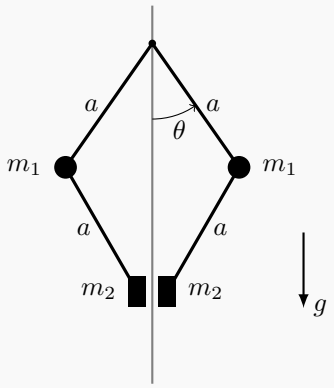
After dropping the total time derivative, we can finally get the Lagrangian

Answer to Problem 3 (c).

$$L = \frac{1}{2}ml^2\dot{\phi}^2 + mla\gamma^2\cos\gamma t\cos\phi + mgl\cos\phi$$

Problem 4.

Find the Lagrangian of a simple pendulum of the system below when placed in a uniform gravitational field (acceleration g). The particle m_2 moves on a vertical axis and the whole system rotates about this axis with a constant angular velocity Ω.



Solution: The position of each particle m_1 is best described in cylindrical coordinates, which is

$$\begin{aligned} r_1 &= a\sin\theta \\ \phi_1 &= \phi \\ z_1 &= a\cos\theta, \end{aligned}$$

where ϕ is the angle of rotation of the system about the axis; $\dot{\phi} = \Omega$. The kinetic energy of each particle m_1 is thus

$$\begin{aligned} T_1 &= \frac{1}{2}m_1(\dot{r}_1^2 + r_1^2\dot{\phi}_1^2 + \dot{z}_1^2) \\ &= \frac{1}{2}m_1(a^2\dot{\theta}^2\cos^2\theta + a^2\Omega^2\sin^2\theta + a^2\dot{\theta}^2\sin^2\theta) \\ &= \frac{1}{2}m_1a^2(\dot{\theta}^2 + \Omega^2\sin^2\theta). \end{aligned} \tag{1.5}$$

The potential energy of this particle can be found by using the z component of his position, namely

$$\begin{aligned} V_1 &= -m_1gz_1 \\ &= -m_1ga\cos\theta. \end{aligned} \tag{1.6}$$

The particle m_2, in its case, can only move up and down, thus its position can be completely defined by

$$z_2 = 2a\cos\theta.$$

Then, the kinetic energy of each particle m_2 is

$$\begin{aligned} T_2 &= \frac{1}{2}m_2\dot{z}_2^2 \\ &= m_2a^2\dot{\theta}^2\sin^2\theta \end{aligned} \tag{1.7}$$

and the potential energy is

$$\begin{aligned} V_2 &= -m_2gz_2 \\ &= -2m_2ga\cos\theta. \end{aligned} \tag{1.8}$$

Using Eq. (1.5), Eq. (1.6), Eq. (1.7) and Eq. (1.8), the Lagrangian of the system is

$$L = 2(T_1 + T_2 - V_1 - V_2),$$

therefore

Answer to Problem 4.

$$L = m_1a^2(\dot{\theta}^2 + \Omega^2\sin^2\theta) + 2m_2a^2\dot{\theta}^2\sin^2\theta + 2(m_1 + 2m_2)ga\cos\theta$$

Chapter 2.

Conservation Laws

Problem 1.

A particle of mass m moving with velocity $\mathbf{v}_1$ leaves a half-space in which its potential energy is a constant U_1 and enters another in which its potential energy is a different constant U_2. Determine the change in the direction of motion of the particle.

Solution: The potential energy only depend on the co-ordinate perpendicular to the plane separating the half-space. Thus, the component of the momentum in that plane is conserved. Denoting by θ_1 and θ_2 the angles between the normal to the plane and the velocities v_1 and v_2 of the particle before and after passing the plane, we have

$$\begin{aligned} & P_1 \sin\theta_1 = P_2 \sin\theta_2 \\ \Rightarrow & v_1 \sin\theta_1 = v_2 \sin\theta_2 \\ \Rightarrow & \frac{\sin\theta_1}{\sin\theta_2} = \frac{v_2}{v_1}. \end{aligned} \tag{2.1}$$

The potential energy of the system is also independent of time, therefore the energy of the particle is conserved. Posing $E_1 = T_1 + U_1$ and $E_2 = T_2 + U_2$ as the energy of the particle before and after passing the plane, the law of conservation of energy requires

$$\begin{aligned} & E_1 = E_2 \\ \Rightarrow & T_1 + U_1 = T_2 + U_2 \\ \Rightarrow & \frac{1}{2}mv_1^2 + U_1 = \frac{1}{2}mv_2^2 + U_2 \\ \Rightarrow & v_2^2 = v_1^2 + \frac{2}{m}(U_1 - U_2). \end{aligned} \tag{2.2}$$

By substituting Eq. (2.2) in the square of Eq. (2.1), we get

$$\left(\frac{\sin\theta_1}{\sin\theta_2}\right)^2 = \frac{v_1^2 + \frac{2}{m}(U_1 - U_2)}{v_1^2}.$$

After taking the square root, the result is

Answer to Problem 1.

$$\frac{\sin\theta_1}{\sin\theta_2} = \sqrt{1 + \frac{2}{mv_1^2}(U_1 - U_2)}$$

Problem 2.

Find the law of transformation of the action S from one inertial frame to another.

Solution: The Lagrangian L and L' of a mechanical system in two inertial frames of reference K and K' are respectively

$$L = T - U = \frac{1}{2}\sum_a m_a v_a^2 - U$$

and

$$L' = T' - U = \frac{1}{2}\sum_a m_a v_a'^2 - U.$$

If the frame K' moves with velocity $\mathbf{V}$ relative to the frame K, the velocities of the particles of the mechanical system relative to the two frames are related by $\mathbf{v}_a = \mathbf{v}'_a + \mathbf{V}$. We can now express the relation of the Lagrangian of the system in the two frames by

$$\begin{aligned} L &= \frac{1}{2}\sum_a m_a(\mathbf{v}'_a + \mathbf{V})^2 - U \\ &= \frac{1}{2}V^2\sum_a m_a + \mathbf{V}\cdot\sum_a m_a\mathbf{v}'_a + \frac{1}{2}\sum_a m_a v_a'^2 - U \\ &= L' + \mathbf{V}\cdot\mathbf{P}' + \frac{1}{2}\mu V^2. \end{aligned} \tag{2.3}$$

Integrating Eq. (2.3) with respect to time, we obtain

$$\begin{aligned} S &= \int L dt \\ &= S' + \int (\mathbf{V}\cdot\mathbf{P}')\, dt + \frac{1}{2}\mu V^2 t \\ &= S' + \mathbf{V}\cdot\int\sum_a m_a\mathbf{v}'_a dt + \frac{1}{2}\mu V^2 t \\ &= S' + \mathbf{V}\cdot\sum_a m_a\mathbf{r}'_a + \frac{1}{2}\mu V^2 t. \end{aligned}$$

The law of transformation of the action S is then

Answer to Problem 2.

$$S = S' + \mu\mathbf{V}\cdot\mathbf{R}' + \frac{1}{2}\mu V^2 t$$

Problem 3.

Obtain the expressions for the Cartesian components and the magnitude of the angular momentum of a particle in cylindrical co-ordinates r, ϕ, z.

Solution: The Cartesian components of the angular momentum are simply

$$\mathbf{M} = \mathbf{r} \times \mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \times m \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = m \begin{bmatrix} y\dot{z} - z\dot{y} \\ z\dot{x} - x\dot{z} \\ x\dot{y} - y\dot{x} \end{bmatrix}. \tag{2.4}$$

In cylindrical co-ordinates, the Cartesian components are expressed by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r\cos\phi \\ r\sin\phi \\ z \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{r}\cos\phi - r\dot{\phi}\sin\phi \\ \dot{r}\sin\phi + r\dot{\phi}\cos\phi \\ \dot{z} \end{bmatrix}.$$

By substituting those in Eq. (2.4), we get

$$\begin{aligned} \mathbf{M} &= m \begin{bmatrix} y\dot{z} - z\dot{y} \\ z\dot{x} - x\dot{z} \\ x\dot{y} - y\dot{x} \end{bmatrix} \\ &= m \begin{bmatrix} (r\dot{z} - z\dot{r})\sin\phi - rz\dot{\phi}\cos\phi \\ -(r\dot{z} - z\dot{r})\cos\phi - rz\dot{\phi}\sin\phi \\ r^2\dot{\phi} \end{bmatrix}. \end{aligned} \tag{2.5}$$

By taking the magnitude of Eq. (2.5), we finally get

Answer to Problem 3.

$$\begin{aligned} M_x &= m(r\dot{z} - z\dot{r})\sin\phi - mrz\dot{\phi}\cos\phi \\ M_y &= -m(r\dot{z} - z\dot{r})\cos\phi - mrz\dot{\phi}\sin\phi \\ M_z &= mr^2\dot{\phi} \\ M^2 &= m^2r^2\dot{\phi}^2(r^2 + z^2) + m^2(r\dot{z} - z\dot{r})^2 \end{aligned}$$

Problem 4.

Obtain the expressions for the Cartesian components and the magnitude of the angular momentum of a particle in spherical co-ordinates r, θ, ϕ.

Solution: In spherical co-ordinates, the Cartesian components are expressed by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} r\sin\phi\cos\theta \\ r\sin\phi\sin\theta \\ r\cos\phi \end{bmatrix}$$

and

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{bmatrix} = \begin{bmatrix} \dot{r}\sin\phi\cos\theta + r\dot{\phi}\cos\phi\cos\theta - r\dot{\theta}\sin\phi\sin\theta \\ \dot{r}\sin\phi\sin\theta + r\dot{\phi}\cos\phi\sin\theta + r\dot{\theta}\sin\phi\cos\theta \\ \dot{r}\cos\phi + r\sin\phi \end{bmatrix}.$$

By substituting those in Eq. (2.4), we get

$$\begin{aligned} \mathbf{M} &= m \begin{bmatrix} y\dot{z} - z\dot{y} \\ z\dot{x} - x\dot{z} \\ x\dot{y} - y\dot{x} \end{bmatrix} \\ &= m \begin{bmatrix} -r^2(\dot{\theta}\sin\phi + \dot{\phi}\sin\theta\cos\theta\cos\phi) \\ r^2(\dot{\theta}\cos\phi - \dot{\phi}\sin\theta\cos\theta\sin\phi) \\ r^2\dot{\phi}\sin^2\theta \end{bmatrix}. \end{aligned} \tag{2.6}$$

By taking the magnitude of Eq. (2.6), we finally get

Answer to Problem 4.

$$\begin{aligned} M_x &= -mr^2(\dot{\theta}\sin\phi + \dot{\phi}\sin\theta\cos\theta\cos\phi) \\ M_y &= mr^2(\dot{\theta}\cos\phi - \dot{\phi}\sin\theta\cos\theta\sin\phi) \\ M_z &= mr^2\dot{\phi}\sin^2\theta \\ M^2 &= m^2r^4(\dot{\theta}^2 + \dot{\phi}^2\sin^2\theta) \end{aligned}$$

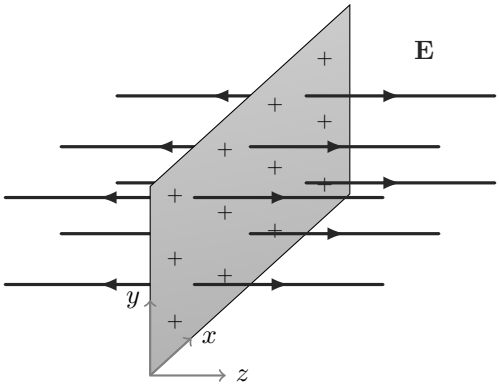
Problem 5.

Which components of momentum **P** and angular momentum **M** are conserved in motion in the following fields ?

(a)

The field of an infinite homogeneous plane.

Solution: solution



Answer to Problem 5 (a).

answer

(b)

The field of an infinite homogeneous cylinder.

Solution: solution

Answer to Problem 5 (b).

answer

(c)

The field of an infinite homogeneous prism.

Solution: solution

Answer to Problem 5 (c).

answer

(d)

The field of two points.

Solution: solution

Answer to Problem 5 (d).

answer

(e)

The field of an infinite homogeneous half-plane.

Solution: solution

Answer to Problem 5 (e).

answer

(f)

The field of a homogeneous cone.

Solution: solution

Answer to Problem 5 (f).

answer

(g)

The field of a homogeneous circular torus.

Solution: solution

Answer to Problem 5 (g).

answer

(h)

The field of an infinite homogeneous cylindrical helix.

Solution: solution

Answer to Problem 5 (h).

answer

Problem 6.

Find the ratio of the times in the same path for particles having different masses but the same potential energy.

Solution: If the two particles have the same path, then the ratio of linear dimension is

$$\frac{l'}{l} = \alpha = 1. \tag{2.7}$$

We can define the ratio of time and mass by

$$\frac{t'}{t} = \beta$$

and

$$\frac{m'}{m} = \gamma.$$

Then, the ratio of kinetic energy is

$$\frac{T'}{T} = \frac{m'\mathbf{v}'}{m\mathbf{v}} = \frac{m'}{m}\left(\frac{\mathbf{dr}'}{\mathbf{dr}}\frac{dt}{dt'}\right)^2 = \frac{\gamma\alpha^2}{\beta^2}.$$

To leave the equation of motion unaltered, the ratio of the kinetic energy and the potential energy must be the same, *i.e.* ,

$$\begin{aligned}\frac{U'}{U} &= \frac{T'}{T} \\ \Rightarrow \alpha^k &= \frac{\gamma\alpha^2}{\beta^2}.\end{aligned}$$

Using Eq. (2.7), we get

$$\begin{aligned}1 &= \frac{\gamma}{\beta^2} \\ \Rightarrow \beta &= \sqrt{\gamma}.\end{aligned}$$

The ratio of the times is then

Answer to Problem 6.

$$\frac{t'}{t} = \sqrt{\frac{m'}{m}}$$

Problem 7.

Find the ratio of the times in the same path for particles the same mass but potential energy differing by a constant factor.

Solution: If the two particles have the same path, then the ratio of linear dimension is

$$\frac{l'}{l} = \alpha = 1. \tag{2.8}$$

We can define the ratio of time by

$$\frac{t'}{t} = \beta.$$

Then, the ratio of kinetic energy is

$$\frac{T'}{T} = \frac{\alpha^2}{\beta^2}.$$

The potential energy of the two particles differ by a constant factor, which mean that

$$\frac{U'}{U} = \gamma.$$

To leave the equation of motion unaltered, the ratio of the kinetic energy and the potential energy must be the same, *i.e.* ,

$$\begin{aligned} \frac{U'}{U} &= \frac{T'}{T} \\ \Rightarrow \gamma &= \frac{\alpha^2}{\beta^2}. \end{aligned}$$

Using Eq. (2.8), we get

$$\begin{aligned} \gamma &= \frac{1}{\beta^2} \\ \Rightarrow \beta &= \sqrt{\frac{1}{\gamma}}. \end{aligned}$$

The ratio of the times is then

Answer to Problem 7.

$$\frac{t'}{t} = \sqrt{\frac{U}{U'}}$$

Chapter 3.

Integration of the Equations of Motion

Problem 1.

Determine the period of oscillations of a simple pendulum (a particle of mass m suspended by a string of length l in a gravitational field) as a function of the amplitude of the oscillations.

Solution: The potential energy of a simple pendulum is

$$U(\phi) = -mgl\cos\phi,$$

see Chap. 1 Prob. 1. If we define ϕ_0 as the maximum value of ϕ, the potential energy is equal to the total energy at this point, that is

$$U(\phi_0) = E = -mgl\cos\phi_0.$$

The energy of the pendulum could also be write using the kinetic energy,

$$E = \frac{1}{2}ml^2\dot{\phi}^2 + U(\phi).$$

This first order differential equation can be integrate to give the the period of oscillation

$$\begin{aligned} T &= 2\sqrt{\frac{ml^2}{2}}\int_{-\phi_0}^{\phi_0}\frac{d\phi}{\sqrt{E-U(\phi)}} \\ &= 4\sqrt{\frac{ml^2}{2}}\int_{0}^{\phi_0}\frac{d\phi}{\sqrt{mgl\cos\phi - mgl\cos\phi_0}} \\ &= 4\sqrt{\frac{l}{2g}}\int_{0}^{\phi_0}\frac{d\phi}{\sqrt{\cos\phi-\cos\phi_0}}. \end{aligned}$$

To solve this integral, we need first to use a trigonometric identity (I'll let you find or derive this identity), giving us

$$T = 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sqrt{\sin^2\frac{\phi_0}{2}-\sin^2\frac{\phi}{2}}}.$$

Next, we use the substitution

$$\sin\xi = \frac{\sin\frac{\phi}{2}}{\sin\frac{\phi_0}{2}}.$$

It's differential form is

$$\begin{aligned} \frac{d\phi}{d\xi} &= \frac{d}{d\xi}\left(2\arcsin\left(\sin\frac{\phi_0}{2}\sin\xi\right)\right) \\ &= \frac{2\sin\frac{\phi_0}{2}\cos\xi}{\sqrt{1-\sin^2\frac{\phi_0}{2}\sin^2\xi}} \end{aligned}$$

The substitution, finally, give

$$\begin{aligned} T &= 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sqrt{\sin^2\frac{\phi_0}{2}-\sin^2\xi\sin^2\frac{\phi_0}{2}}} \\ &= 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sin\frac{\phi_0}{2}\sqrt{1-\sin^2\xi}} \\ &= 2\sqrt{\frac{l}{g}}\int_0^{\phi_0}\frac{d\phi}{\sin\frac{\phi_0}{2}\cos\xi} \\ &= 2\sqrt{\frac{l}{g}}\int_{\arcsin(0)}^{\arcsin(1)}\frac{2\sin\frac{\phi_0}{2}\cos\xi d\xi}{\sin\frac{\phi_0}{2}\cos\xi\sqrt{1-\sin^2\frac{\phi_0}{2}\sin^2\xi}} \\ &= 4\sqrt{\frac{l}{g}}\int_0^{\frac{\pi}{2}}\frac{d\xi}{\sqrt{1-\sin^2\frac{\phi_0}{2}\sin^2\xi}}. \end{aligned}$$

We can also write

$$T = 4\sqrt{\frac{l}{g}} K\left(\sin\frac{\phi_0}{2}\right),$$

where

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{d\xi}{\sqrt{1 - k\sin^2\xi}}. \tag{3.1}$$

The integral K is known as the complete elliptic integral of the first kind. To solve this integral, we first need to see that the integrand is of the form

$$f(x) = (1-x)^{-\frac{1}{2}}.$$

The n derivative of this function is

$$f^{(n)}(x) = \frac{(2n-1)!!}{2^n}(1-x)^{\frac{-2n+1}{2}},$$

thus the Maclaurin Series is

$$f(x) = \sum_{n\geq 0} \frac{(2n-1)!!}{2^n \cdot n!} x^n.$$

Using this result in Eq. (3.1), we get

$$\begin{aligned} K(k) &= \int_0^{\frac{\pi}{2}} \sum_{n\geq 0} \frac{(2n-1)!!}{2^n \cdot n!} k^{2n} \sin^{2n}\xi \, d\xi \\ &= \sum_{n\geq 0} \frac{(2n-1)!!}{2^n \cdot n!} k^{2n} \int_0^{\frac{\pi}{2}} \sin^{2n}\xi \, d\xi. \end{aligned}$$

This last integral can be solve using the beta function

$$\mathcal{B}(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = 2\int_0^{\frac{\pi}{2}} \sin^{2x-1} t \cos^{2y-1} t \, dt,$$

with $x = \frac{2n+1}{2}$ and $y = \frac{1}{2}$. Thus, the beta function become

$$\begin{aligned} \frac{1}{2}\mathcal{B}(x,y) &= \frac{\Gamma(x)\Gamma(y)}{2\Gamma(x+y)} \\ &= \frac{\sqrt{\pi}}{2}\frac{\Gamma(\frac{2n+1}{2})}{n!} \\ &= \frac{\pi}{2}\frac{(2n+1)!!}{2^n \cdot n!} \end{aligned}$$

The elliptic integral is thereby

$$\begin{aligned} K(k) &= \frac{\pi}{2}\sum_{n\geq 0}\left[\frac{(2n-1)!!}{2^n \cdot n!} k^n\right]^2 \\ &= \frac{\pi}{2}\sum_{n\geq 0}\left[\frac{(2n-1)!!}{(2n)!!} k^n\right]^2. \end{aligned}$$

The period of a simple pendulum is

$$T = 2\pi\sqrt{\frac{l}{g}}\sum_{n\geq 0}\left[\frac{(2n-1)!!}{(2n)!!}\sin^n\frac{\phi_0}{2}\right]^2$$

By expanding the sum, we get

$$T = 2\pi\sqrt{\frac{l}{g}}\left(1 + \left(\frac{1}{2}\right)^2 \sin^2\frac{\phi}{2} + \left(\frac{1\cdot 3}{2\cdot 4}\right)^2 \sin^4\frac{\phi}{2} + \left(\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\right)^2 \sin^6\frac{\phi}{2} + \ldots\right).$$

Using the Maclaurin series

$$\sin\frac{\phi_0}{2}=\frac{1}{2}\phi_0-\frac{1}{48}\phi_0^3+\frac{1}{3840}\phi_0^5-\frac{1}{645120}\phi_0^7+\ldots,$$

we finally get

Answer to Problem 1.

$$T=2\pi\sqrt{\frac{l}{g}}\left(1+\frac{1}{16}\phi_0^2+\frac{11}{3072}\phi_0^4+\ldots\right)$$

Problem 2.

Determine the period of oscillation, as a function of the energy, when a particle of mass m moves in fields for which the potential energy is

(a)

$$U = A|x|^n$$

Solution: The total energy of the particle is

$$E = \frac{1}{2}m\dot{x}^2 + U(x).$$

Knowing that the maximum value of E is at x_1, this position is

$$\begin{aligned} E = U(x_1) &= A|x_1|^n \\ \Rightarrow |x_1| &= \left(\frac{E}{A}\right)^{1/n}, \\ \Rightarrow x_1 &= \pm\left(\frac{E}{A}\right)^{1/n}, \end{aligned}$$

we get the period of oscillation,

$$\begin{aligned} T &= 2\sqrt{\frac{m}{2}}\int_{-\left(\frac{E}{A}\right)^{1/n}}^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{E - A|x|^n}} \\ &= 2\sqrt{2m}\int_0^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{E - A|x|^n}}. \end{aligned}$$

It is possible to remove the absolute value, because the integral is over positive x, thus

$$\begin{aligned} T &= 2\sqrt{2m}\int_0^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{E - Ax^n}} \\ &= 2\sqrt{\frac{2m}{E}}\int_0^{\left(\frac{E}{A}\right)^{1/n}} \frac{dx}{\sqrt{1 - \frac{A}{E}x^n}}. \end{aligned}$$

Using the substitution $y = \left(\frac{A}{E}\right)^{\frac{1}{n}} x$, we get

$$\begin{aligned} T &= 2\sqrt{\frac{2m}{E}}\int_0^1 \frac{\left(\frac{A}{E}\right)^{\frac{1}{n}} dy}{\sqrt{1 - \frac{A}{E}\left(\left(\frac{A}{E}\right)^{\frac{-1}{n}} y\right)^n}} \\ &= 2\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}}\int_0^1 \frac{dy}{\sqrt{1 - y^n}}. \end{aligned}$$

Using another substitution $u = y^n$, we get

$$T = 2\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}}\int_0^1 \frac{u^{\frac{1}{n}}\,du}{nu\sqrt{1-u}}.$$

It is possible to express this integral in term of the beta function

$$\mathcal{B}(z_1, z_2) = \int_0^1 t^{z_1 - 1}(1-t)^{z_2 - 1}\,dt = \frac{\Gamma(z_1)\,\Gamma(z_2)}{\Gamma(z_1 + z_2)},$$

with $z_1 = \frac{1}{n}$ and $z_2 = \frac{1}{2}$. This give us

$$T = \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}} \frac{\Gamma\left(\frac{1}{n}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{n}\right)}.$$

Knowing that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, we finally have

Answer to Problem 2 (a).

$$T = \frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{\frac{1}{n}} \frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{2}+\frac{1}{n}\right)}$$

(b)

$$U = \frac{-U_0}{\cosh^2 \alpha x}$$

Solution: The two boundaries of the energy are at $E = 0 \Rightarrow x = x_1$ and $E = -U_0 \Rightarrow x = 0$. The period of oscillation is thus

$$\begin{aligned} T &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E-U}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E+\frac{U_0}{\cosh^2 \alpha x}}} \\ &= 2\sqrt{2m}\int_0^{x_1} \frac{\cosh^2 \alpha x \quad dx}{\sqrt{E\cosh^2 \alpha x + U_0}}. \end{aligned}$$

Using the identity $\cosh^2 \alpha x = 1 + \sinh^2 \alpha x$, we get

$$T = 2\sqrt{2m}\int_0^{x_1} \frac{\cosh^2 \alpha x \quad dx}{\sqrt{E\left(1+\sinh^2 \alpha x\right) + U_0}}.$$

The substitution $y = \sinh \alpha x$, give us $dy = \alpha \cosh \alpha x dx$ and

$$\begin{aligned} y(x_1) &= \sinh \alpha x \\ &= \sqrt{\sinh^2 \alpha x} \\ &= \sqrt{\cosh^2 \alpha x - 1} \\ &= \sqrt{\frac{-U_0}{E} - 1}. \end{aligned}$$

Using this substitution, we get

$$\begin{aligned} T &= \frac{2}{\alpha}\sqrt{2m}\int_0^{\sqrt{\frac{-U_0}{E}-1}} \frac{dy}{E\left(1+y^2\right)+U_0} \\ &= \frac{2}{\alpha}\sqrt{2m}\int_0^{\sqrt{\frac{-U_0}{E}-1}} \frac{dy}{E+Ey^2-E\cosh^2 \alpha x} \end{aligned}$$

Factoring $|E|$ out of the square root (because $E < 0$), we have

$$\begin{aligned} T &= \frac{2}{\alpha}\sqrt{\frac{2m}{|E|}}\int_0^{\sqrt{\frac{-U_0}{E}-1}} \frac{dy}{1-\cosh^2 \alpha x + y^2} \\ &= \frac{2}{\alpha}\sqrt{\frac{2m}{|E|}}\int_0^{\sqrt{\frac{-U_0}{E}-1}} \frac{dy}{\sqrt{\frac{-U_0}{E}-1+y^2}} \end{aligned}$$

This integral is of the form

$$\int_0^a \frac{dz}{\sqrt{a+z^2}} = \sin^{-1}\left(\frac{z}{a}\right) = \frac{\pi}{2}.$$

Thus, the period of oscillation is

Answer to Problem 2 (b).

$$T = \frac{\pi}{\alpha}\sqrt{\frac{2m}{|E|}}$$

(c)

$$U = U_0 \tan^2 \alpha x$$

Solution: The period of oscillation is

$$\begin{aligned}
T &= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E-U}} \\
&= 2\sqrt{2m}\int_0^{x_1} \frac{dx}{\sqrt{E-U_0\tan^2\alpha x}} \\
&= 2\sqrt{2m}\int_0^{x_1} \frac{\cos\alpha x \quad dx}{\sqrt{E\cos^2\alpha x - U_0\sin^2\alpha x}} \\
&= 2\sqrt{2m}\int_0^{x_1} \frac{\cos\alpha x \quad dx}{\sqrt{E-(E+U_0)\sin^2\alpha x}}.
\end{aligned}$$

Using the substitution $y = i\sin\alpha x$ ($dy = i\alpha\cos\alpha x$), we get

$$\begin{aligned}
T &= \frac{2\sqrt{2m}}{i\alpha}\int_0^{i\sin\alpha x_1} \frac{dy}{\sqrt{E+(E+U_0)\,y^2}} \\
&= \frac{2}{i\alpha}\sqrt{\frac{2m}{E+U_0}}\int_0^{i\sin\alpha x_1} \frac{dy}{\sqrt{\frac{E}{(E+U_0)}+y^2}} \\
&= \frac{2}{i\alpha}\sqrt{\frac{2m}{E+U_0}}\sin^{-1}\left(\frac{i\sin\alpha x_1}{\sqrt{\frac{E}{(E+U_0)}}}\right)
\end{aligned}$$

Knowing

$$x_1 = \frac{1}{\alpha}\tan^{-1}\sqrt{\frac{E}{U_0}},$$

we finally get

$$\begin{aligned}
T &= \frac{2}{i\alpha}\sqrt{\frac{2m}{E+U_0}}\sin^{-1}\left(\frac{i\sin\left(\tan^{-1}\sqrt{\frac{E}{U_0}}\right)}{\sqrt{\frac{E}{(E+U_0)}}}\right) \\
&= \frac{2}{i\alpha}\sqrt{\frac{2m}{E+U_0}}\sin^{-1}\left(\frac{i\sqrt{\frac{E}{U_0}}}{\sqrt{1+\frac{E}{U_0}}\sqrt{\frac{E}{(E+U_0)}}}\right) \\
&= \frac{2}{i\alpha}\sqrt{\frac{2m}{E+U_0}}\sin^{-1} i \\
&= \frac{2}{i\alpha}\sqrt{\frac{2m}{E+U_0}}\frac{\pi i}{2}.
\end{aligned}$$

Thus, the period of oscillation is

Answer to Problem 2 (c).

$$T = \frac{\pi}{\alpha}\sqrt{\frac{2m}{E+U_0}}$$

Problem 3.

A system consists of one particle of mass M and n particles with equal masses m. Eliminate the motion of the centre of mass and so reduce the problem to one involving n particles.

Solution: Let $\mathbf{R}$ be the radius vector of the particle of mass M, and $\mathbf{R}_a\,(a = 1, 2, \ldots, n)$ those of the particles of mass m. We put $\mathbf{r}_a \equiv \mathbf{R}_a - \mathbf{R}$ and take the origin to be at the centre of mass, namely

$$M\mathbf{R} + m\sum_a \mathbf{R}_a = 0.$$

Thus, we got

$$\mathbf{R} = -\frac{m}{M}\left(\sum_a \mathbf{r}_a + n\mathbf{R}\right) = -\frac{m}{\mu}\sum_a \mathbf{r}_a, \tag{3.2}$$

where $\mu \equiv M + nm$. The Lagrangian of the system is

$$L = \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}m\sum_a \dot{\mathbf{R}}_a^2 - U.$$

If we substitute Eq. (3.2) and $\mathbf{R}_a = \mathbf{R} + \mathbf{r}_a$, we get

$$\begin{aligned} L &= \frac{1}{2}\frac{Mm^2}{\mu^2}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 + \frac{1}{2}nm\dot{\mathbf{R}}^2 + \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 + m\dot{\mathbf{R}}\sum_a \dot{\mathbf{r}}_a - U \\ &= \frac{1}{2}\frac{Mm^2}{\mu^2}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 + \frac{1}{2}\frac{m^3}{\mu^2}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 + \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{m^2}{\mu}\left(\sum_a \dot{\mathbf{r}}_a\right)^2 - U \\ &= \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{1}{2}\frac{m^2}{\mu}\left(\frac{2\mu - M - nm}{\mu}\right)\left(\sum_a \dot{\mathbf{r}}_a\right)^2. \end{aligned}$$

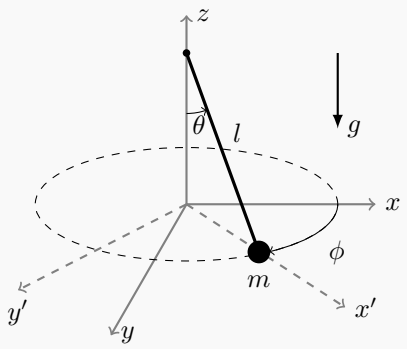
The Lagrangian is thus only a function of $\mathbf{r}_a$ and it's time derivative, which is

Answer to Problem 3.

$$L = \frac{1}{2}m\sum_a \dot{\mathbf{r}}_a^2 - \frac{1}{2}\frac{m^2}{\mu}\left(\sum_a \dot{\mathbf{r}}_a\right)^2$$

Problem 4.

Integrate the equations of motion for a spherical pendulum (a particle of mass m moving on the surface of a sphere of radius l in a gravitational field).



Solution: The Cartesian position of the particle m is

$$\begin{aligned} x &= l\sin\theta\cos\phi \\ y &= l\sin\theta\sin\phi \\ z &= -l\cos\theta. \end{aligned}$$

The time derivatives of these coordinates are

$$\begin{aligned} \dot{x} &= l\dot{\theta}\cos\theta\cos\phi - l\dot{\phi}\sin\theta\sin\phi \\ \dot{y} &= l\dot{\theta}\cos\theta\sin\phi + l\dot{\phi}\sin\theta\cos\phi \\ \dot{z} &= l\dot{\theta}\sin\theta. \end{aligned}$$

Substituting these expressions in the Lagrangian, we obtain the desired Lagrangian form, that is

$$\begin{aligned} L &= \frac{1}{2}m\left(\dot{x}^2+\dot{y}^2+\dot{z}^2\right) - mgz \\ &= \frac{1}{2}ml^2\left(\dot{\theta}^2+\dot{\phi}^2\sin^2\theta\right) + mgl\cos\theta. \end{aligned}$$

The Lagrangian does not involve ϕ explicitly, thus this co-ordinate is cyclic and the generalized momentum p_ϕ is conserved. This momentum is the same as the z-component of angular momentum M_z and is written

$$M_z = p_\phi = \frac{\partial L}{\partial\dot{\phi}} = ml^2\dot{\phi}\sin^2\theta. \tag{3.3}$$

The energy of the pendulum is

$$E = \frac{1}{2}ml^2\left(\dot{\theta}^2+\dot{\phi}^2\sin^2\theta\right) - mgl\cos\theta.$$

Substituting $\dot{\phi}$ with Eq. (3.3) in the energy we get

$$\begin{aligned} E &= \frac{1}{2}ml^2\dot{\theta}^2 + \frac{M_z^2}{2ml^2\sin^2\theta} - mgl\cos\theta \\ &= \frac{1}{2}ml^2\dot{\theta}^2 + U_{\text{eff}}(\theta), \end{aligned}$$

where $U_{\text{eff}}(\theta)$ is the effective potential energy. Notice that the energy was re-written as a function of only one co-ordinate (*i.e.* θ). This is equivalent to the one particle problem we have already see, thus

$$\begin{aligned} \frac{d\theta}{dt} &= \sqrt{\frac{2\left(E-U_{\text{eff}}(\theta)\right)}{ml^2}} \\ \Rightarrow t &= \int\frac{d\theta}{\sqrt{2\left(E-U_{\text{eff}}(\theta)\right)}}. \end{aligned}$$

This integral lead to an elliptic integral of the first kind see Chap. 3 Prob. 1. We also need to find the solution to the angle ϕ. To do so, we can use the Eq. (3.3) with the chain rule from calculus, that is

$$\begin{aligned}\frac{M_z}{ml^2\sin^2\theta} &= \frac{d\phi}{dt}\\ &= \frac{d\phi}{d\theta}\frac{d\theta}{dt}\\ &= \frac{d\phi}{d\theta}\sqrt{\frac{2\left(E-U_{\text{eff}}(\theta)\right)}{ml^2}}.\end{aligned}$$

Finally, we get

$$\begin{aligned}\frac{d\phi}{d\theta} &= \frac{M_z}{l\sin^2\theta}\sqrt{\frac{1}{2m\left(E-U_{\text{eff}}(\theta)\right)}}\\ \Rightarrow\phi &= \frac{M_z}{l}\sqrt{\frac{1}{2m}}\int\frac{d\theta}{\sin^2\theta\left(E-U_{\text{eff}}(\theta)\right)}.\end{aligned}$$

This integral lead to an elliptic integral of the third kind (I really don't want to solve this). It is possible to find the range of angle θ. At the maximum and minimum value of θ, the pendulum as no kinetic energy, thus

$$\begin{aligned}E &= U_{\text{eff}}\\ &= \frac{M_z^2}{2ml^2\sin^2\theta} - mgl\cos\theta\\ &= \frac{M_z^2}{2ml^2\left(1-\cos^2\theta\right)} - mgl\cos\theta\end{aligned}$$

We can find a cubic algebraic equation for $\cos\theta$, that is

$$\begin{aligned}0 &= \left(E+mgl\cos\theta\right)\left(2ml^2\left(1-\cos^2\theta\right)\right) - M_z^2\\ &= mgl\left(\cos^3\theta-\cos\theta\right) + E\left(\cos^2\theta-1\right) + \frac{M_z^2}{2ml^2}.\end{aligned}$$

The centrifugal part of the effective potential U_{eff}, namely

$$\frac{M_z^2}{2ml^2\sin^2\theta},$$

must be positive. As we can see, this term diverge at $\theta\to 0$ and $\theta\to\pi$, thus the angle θ is bound. This can be write as

$$\theta\in[\theta_p,\theta_a] \qquad\qquad 0<\theta_p\leq\theta_a<\pi,$$

where the subscripts of the angles θ_p and θ_a are for the perigee and apogee. This mean that the two roots of the cubic algebraic equation for $\cos\theta$ is bound between -1 and $+1$. To summarize, the solution of the equation of motion of the spherical pendulum is

Answer to Problem 4.

$$t = \int_{\theta_p}^{\theta_a}\frac{d\theta}{\sqrt{2\left(E-U_{\text{eff}}(\theta)\right)}} \qquad\qquad \phi = \frac{M_z}{l}\sqrt{\frac{1}{2m}}\int_{\theta_p}^{\theta_a}\frac{d\theta}{\sin^2\theta\left(E-U_{\text{eff}}(\theta)\right)}$$

Problem 5.

Integrate the equations of motion for a particle moving on the surface of a cone (of vertical axis 2α) placed vertically and with vertex downwards in a gravitational field.

Solution: By using the spherical co-ordinate, we can write the position of the particle as

$$\begin{aligned} x &= r\cos\phi\sin\alpha \\ y &= r\sin\phi\sin\alpha \\ z &= r\cos\alpha. \end{aligned}$$

By taking the time derivative of those, we obtain

$$\begin{aligned} \dot{x} &= \dot{r}\cos\phi\sin\alpha - r\dot{\phi}\sin\phi\sin\alpha \\ \dot{y} &= \dot{r}\sin\phi\sin\alpha + r\dot{\phi}\cos\phi\sin\alpha \\ \dot{z} &= \dot{r}\cos\alpha. \end{aligned}$$

The Lagrangian is thus

$$\begin{aligned} L &= \frac{1}{2}m\left(\dot{x}^2+\dot{y}^2+\dot{z}^2\right) - mgz \\ &= \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\sin^2\alpha\right) - mgr\cos\alpha. \end{aligned}$$

The co-ordinate ϕ is cyclic, thus

$$M_z = p_\phi = \frac{\partial L}{\partial\dot{\phi}} = mr^2\dot{\phi}\sin^2\alpha \tag{3.4}$$

is conserved. The energy of the particle is

$$E = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\sin^2\alpha\right) + mgr\cos\alpha. \tag{3.5}$$

We can rewrite Eq. (3.4) as

$$\dot{\phi}^2 = \frac{M_z^2}{m^2r^4\sin^4\alpha}$$

and substituting in Eq. (3.5), we get

$$\begin{aligned} E &= \frac{1}{2}m\dot{r}^2 + \frac{M_z^2}{2mr^2\sin^2\alpha} + mgr\cos\alpha \\ &= \frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r). \end{aligned}$$

Hence,

$$t = \sqrt{\frac{m}{2}}\int\frac{dr}{\sqrt{E-U_{\text{eff}}(r)}}.$$

From Eq. (3.4), it is possible to write

$$\begin{aligned} &\frac{d\phi}{dt} = \frac{M_z}{mr^2\sin^2\alpha} \\ \Rightarrow &\frac{d\phi}{dr}\frac{dr}{dt} = \frac{M_z}{mr^2\sin^2\alpha} \\ \Rightarrow &\frac{d\phi}{dr}\sqrt{\frac{2\left(E-U_{\text{eff}}(r)\right)}{m}} = \frac{M_z}{mr^2\sin^2\alpha} \\ \Rightarrow &\phi = \frac{M_z}{\sqrt{2m}\sin^2\alpha}\int\frac{dr}{r^2\sqrt{E-U_{\text{eff}}(r)}}. \end{aligned}$$

It is possible to find the range of r. At the maximum and minimum value of r, the particle as no kinetic energy, thus

$$\begin{aligned} E &= U_{\text{eff}} \\ &= \frac{M_z^2}{2mr^2 \sin^2 \alpha} + mgr \cos \alpha \end{aligned}$$

We can find a cubic algebraic equation for r, that is

$$\begin{aligned} 0 &= r^2 \left(E - mgr \cos \alpha \right) - \frac{M_z^2}{2m \sin^2 \alpha} \\ &= mgr^3 \cos \alpha - Er^2 + \frac{M_z^2}{2m \sin^2 \alpha}. \end{aligned}$$

This equation as two positive roots, r_p and r_a, which are the turning points of the motion. To summarize, the solution of the equation of motion for a particle moving on the surface of a cone is

Answer to Problem 5.

$$t = \sqrt{\frac{m}{2}} \int_{r_p}^{r_a} \frac{dr}{\sqrt{E - U_{\text{eff}}(r)}} \qquad \phi = \frac{M_z}{\sqrt{2m} \sin^2 \alpha} \int_{r_p}^{r_a} \frac{dr}{r^2 \sqrt{E - U_{\text{eff}}(r)}}$$

Problem 6.

Integrate the equations of motion for a pendulum of mass m_2, with a mass m_1 at the point of support which can move on a horizontal line lying in the plane which m_2 moves (Chap. 1 Prob. 2).

Solution: From Chap. 1 Prob. 2, the Lagrangian of the system is

$$L = \frac{1}{2}(m_1+m_2)\dot{x}^2 + \frac{1}{2}m_2(l^2\dot{\phi}^2 + 2l\dot{x}\dot{\phi}\cos\phi) + m_2 gl\cos\phi.$$

The co-ordinate x is cyclic, thus

$$p_x = \frac{\partial L}{\partial \dot{x}} = (m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi = \text{constant} \tag{3.6}$$

is conserved. It is always possible to find an inertial frame of reference where $p_x = 0$, using this frame we get

$$\begin{aligned}
&(m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi = 0 \\
\Rightarrow &\int \Big((m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi\Big)\,dt = \text{constant} \\
\Rightarrow &(m_1+m_2)x + m_2 l\sin\phi = (m_1+m_2)R_x = \text{constant}.
\end{aligned} \tag{3.7}$$

This express the fact that the centre of mass $\mathbf{R}$ of the system does not move horizontally. It is also possible to rewrite Eq. (3.6) as

$$\begin{aligned}
(m_1+m_2)\dot{x} + m_2 l\dot{\phi}\cos\phi &= 0 \\
\Rightarrow \dot{x} &= \frac{-m_2 l\dot{\phi}\cos\phi}{m_1+m_2}
\end{aligned}$$

Plugging this into the energy of the system we get

$$\begin{aligned}
E &= \frac{1}{2}(m_1+m_2)\dot{x}^2 + \frac{1}{2}m_2(l^2\dot{\phi}^2 + 2l\dot{x}\dot{\phi}\cos\phi) - m_2 gl\cos\phi \\
&= \frac{m_2^2 l^2\dot{\phi}^2\cos^2\phi}{2(m_1+m_2)} + \frac{1}{2}m_2 l^2\dot{\phi}^2 - \frac{m_2^2 l^2\dot{\phi}^2\cos^2\phi}{m_1+m_2} - m_2 gl\cos\phi \\
&= \frac{1}{2}m_2 l^2\dot{\phi}^2\left(1 - \frac{m_2}{m_1+m_2}\cos^2\phi\right) - m_2 gl\cos\phi.
\end{aligned}$$

Hence,

$$\begin{aligned}
t &= l\sqrt{\frac{m_2}{2}}\int\sqrt{\frac{1-\frac{m_2}{m_1+m_2}\cos^2\phi}{E+m_2 gl\cos\phi}}\,d\phi \\
&= l\sqrt{\frac{m_2}{2(m_1+m_2)}}\int\sqrt{\frac{m_1+m_2\sin^2\phi}{E+m_2 gl\cos\phi}}\,d\phi.
\end{aligned}$$

Using Eq. (3.7), we can express the position of the mass m_2 as

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x + l\sin\phi \\ l\cos\phi \end{bmatrix} = \begin{bmatrix} R_x - l\sin\phi\left(1 - \frac{m_2}{m_1+m_2}\right) \\ l\cos\phi \end{bmatrix}.$$

The path of the particle of mass m_2 is thus an arc of an ellipse center at $(R_x, 0)$ with horizontal semi-axis $lm_1/(m_1+m_2)$ and vertical semi-axis l. It is possible to see that when $m_1 \to \infty$, the path return to the simple pendulum. The solution of the equations of motion is thus

Answer to Problem 6.

$$t = l\sqrt{\frac{m_2}{2}}\int\sqrt{\frac{1-\frac{m_2}{m_1+m_2}\cos^2\phi}{E+m_2 gl\cos\phi}}\,d\phi \qquad \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} R_x - l\sin\phi\left(1 - \frac{m_2}{m_1+m_2}\right) \\ l\cos\phi \end{bmatrix}$$

Problem 7.

Find the time dependence of the co-ordinate of a particle with energy $E = 0$ moving in a parabola in a field $U = -\alpha/r$.

Solution: From the formulae (14.6) of the book, we have the integral

$$\begin{aligned} t &= \int \frac{dr}{\sqrt{\frac{2}{m}\left(E - U(r)\right) - \frac{M^2}{m^2 r^2}}} \\ &= \int \frac{dr}{\sqrt{\frac{2\alpha}{mr} - \frac{M^2}{m^2 r^2}}} \\ &= \int \frac{rdr}{\sqrt{\frac{2\alpha}{m}r - \frac{M^2}{m^2}}}. \end{aligned}$$

We use the substitution

$$\begin{aligned} \frac{m}{M}\eta &= \sqrt{\frac{2\alpha}{m}r - \frac{M^2}{m^2}} \\ \Rightarrow r &= \frac{M^2}{2m\alpha}\left(1+\eta^2\right) = \frac{1}{2}p\left(1+\eta^2\right), \end{aligned} \tag{3.8}$$

with the differential form

$$dr = \frac{M^2}{m\alpha}\eta d\eta.$$

Hence, the integral become

$$\begin{aligned} t &= \frac{M^3}{2m\alpha^2}\int \left(1+\eta^2\right) d\eta \\ &= \frac{M^3}{2m\alpha^2}\left(\eta + \frac{1}{3}\eta^3\right) \\ &= \sqrt{\frac{mp^3}{\alpha}}\frac{1}{2}\eta\left(\eta + \frac{1}{3}\eta^3\right). \end{aligned}$$

It is important to specify that the parameter η varies from $-\infty$ to ∞. Using Eq. (3.8) and

$$\cos\phi = \frac{p}{r} - 1,$$

it is possible to find the Cartesian co-ordinates

$$x = r\cos\phi = \frac{1}{2}p\left(1-\eta^2\right)$$

and

$$y = \sqrt{r^2 - x^2} = p\eta.$$

The parametric form of the required dependence are thus

Answer to Problem 7.

$$r = \frac{1}{2}p\left(1+\eta^2\right) \qquad t = \sqrt{\frac{mp^3}{\alpha}}\frac{1}{2}\eta\left(\eta + \frac{1}{3}\eta^3\right)$$

$$x = \frac{1}{2}p\left(1-\eta^2\right) \qquad y = p\eta$$

Problem 8.

Integrate the equations of motion for a particle in a central field

$$U = -\frac{\alpha}{r^2} \qquad (\alpha > 0).$$

Solution: From the formulae (14.6) of the book, we have the integral

$$\begin{aligned} t &= \int \frac{dr}{\sqrt{\frac{2}{m}\left(E - U(r)\right) - \frac{M^2}{m^2 r^2}}} \\ &= \int \frac{dr}{\sqrt{\frac{2E}{m}r^2 + \frac{2\alpha}{m} - \frac{M^2}{m^2}}} \\ &= \sqrt{\frac{m}{2E}} \int \frac{rdr}{\sqrt{r^2 + \left(\frac{\alpha}{E} - \frac{M^2}{2mE}\right)}}. \end{aligned}$$

We use the substitution

$$u = r^2 + \left(\frac{\alpha}{E} - \frac{M^2}{2mE}\right)$$

with the differential form

$$du = 2rdr.$$

Hence, the integral become

$$\begin{aligned} t &= \frac{1}{2}\sqrt{\frac{m}{2E}} \int \frac{du}{\sqrt{u}} \\ &= \sqrt{\frac{m}{2E}}\sqrt{u} \\ &= \sqrt{\frac{m}{2E}}\sqrt{r^2 + \frac{\alpha}{E} - \frac{M^2}{2mE}} \\ &= \frac{1}{E}\sqrt{\frac{m}{2}\left(Er^2 + \alpha - \frac{M^2}{2m}\right)}. \end{aligned}$$

The formulae (14.7) of the book give us the equation of the path

$$\begin{aligned} \phi &= \int \frac{Mdr}{r^2\sqrt{2m\left(E - U(r)\right) - \frac{M^2}{r^2}}} \\ &= \int \frac{Mdr}{r^2\sqrt{2mE + \frac{2m\alpha}{r^2} - \frac{M^2}{r^2}}}. \end{aligned}$$

Using the substitution

$$u = \frac{1}{r}$$

with the differential form

$$du = -\frac{1}{r^2}dr,$$

the integral become

$$\begin{aligned} \phi &= -\int \frac{du}{\sqrt{2mE + \left(2m\alpha - M^2\right)u^2}} \\ &= -\frac{1}{\sqrt{2mE}} \int \frac{du}{\sqrt{1 + (ku)^2}}, \end{aligned} \tag{3.9}$$

with

$$k = \sqrt{\frac{2m\alpha - M^2}{2mE}}. \tag{3.10}$$

From there, the solution must be divided for the three possible cases : **(a)** $E > 0$, $M^2 > 2m\alpha$; **(b)** $E > 0$, $M^2 < 2m\alpha$; **(c)** $E < 0$ $M^2 < 2m\alpha$. It is also interesting to know the the path is a Cotes's spiral[1].

(c) : Eq. (3.10) is still

$$k = \sqrt{\frac{2m\alpha - M^2}{2mE}}.$$

By using the substitution

$$ku = \sinh\theta$$

with the differential form

$$kdu = \cosh\theta d\theta,$$

the integral Eq. (3.9) become

$$\begin{aligned}\phi &= -\frac{1}{\sqrt{2mE}}\int \frac{\cosh\theta d\theta}{k\sqrt{1+\sinh^2\theta}} \\ &= -\frac{1}{\sqrt{2m\alpha - M^2}}\theta \\ &= -\frac{1}{\sqrt{2m\alpha - M^2}}\sinh^{-1}\frac{k}{r}.\end{aligned}$$

The equation of the path for the case (c) is thus

$$\begin{aligned}\frac{1}{r} &= \frac{1}{k}\sinh\left(\phi\sqrt{2m\alpha - M^2}\right) \\ &= \sqrt{\frac{2mE}{2m\alpha - M^2}}\sinh\left(\phi\sqrt{\frac{2m\alpha}{M^2} - 1}\right).\end{aligned}$$

(b) : Eq. (3.10) become

$$k = \frac{\sqrt{2m\alpha - M^2}}{i\sqrt{2m|E|}}.$$

By using the substitution

$$iku = \cosh\theta$$

with the differential form

$$ikdu = \sinh\theta d\theta,$$

the integral Eq. (3.9) become

$$\begin{aligned}\phi &= -\frac{1}{i\sqrt{2m|E|}}\int \frac{\sinh\theta d\theta}{ik\sqrt{1-\cosh^2\theta}} \\ &= \frac{1}{\sqrt{2m\alpha - M^2}}\theta \\ &= \frac{1}{\sqrt{2m\alpha - M^2}}\cosh^{-1}\frac{ik}{r}.\end{aligned}$$

The equation of the path for the case (b) is thus

$$\begin{aligned}\frac{1}{r} &= \frac{1}{ik}\cosh\left(\phi\sqrt{2m\alpha - M^2}\right) \\ &= \sqrt{\frac{2m|E|}{2m\alpha - M^2}}\cosh\left(\phi\sqrt{\frac{2m\alpha}{M^2} - 1}\right).\end{aligned}$$

[1] Whittaker ET (1937). A Treatise on the Analytical Dynamics of Particles and Rigid Bodies, with an Introduction to the Problem of Three Bodies (4th ed.). New York: Dover Publications. pp. 80–83. ISBN 978-0-521-35883-5.

(a) : Eq. (3.10) become

$$k = \frac{i\sqrt{2m\alpha - M^2}}{\sqrt{2mE}}.$$

By using the substitution

$$ku = i\cosh\theta$$

with the differential form

$$kdu = i\sinh\theta d\theta,$$

the integral Eq. (3.9) become

$$\begin{aligned}\phi &= -\frac{1}{\sqrt{2mE}}\int \frac{i\sinh\theta d\theta}{k\sqrt{1-\cosh^2\theta}} \\ &= -\frac{i}{\sqrt{2m\alpha - M^2}}\theta \\ &= -\frac{i}{\sqrt{2m\alpha - M^2}}\cosh^{-1}\frac{k}{ir}.\end{aligned}$$

The equation of the path for the case (b) is thus

$$\begin{aligned}\frac{1}{r} &= \frac{i}{k}\cosh\left(\frac{\phi}{i}\sqrt{2m\alpha - M^2}\right) \\ &= \sqrt{\frac{2mE}{M^2 - 2m\alpha}}\cos\left(\phi\sqrt{1-\frac{2m\alpha}{M^2}}\right).\end{aligned}$$

To summarize the equations of motion for a particle in a central inverse-cube law force is

Answer to Problem 8.

(a) for $E > 0$ and $\dfrac{M^2}{2m} > \alpha$, $\quad \dfrac{1}{r} = \sqrt{\dfrac{2mE}{M^2 - 2m\alpha}}\cos\left(\phi\sqrt{1-\dfrac{2m\alpha}{M^2}}\right)$

(b) for $E > 0$ and $\dfrac{M^2}{2m} < \alpha$, $\quad \dfrac{1}{r} = \sqrt{\dfrac{2mE}{2m\alpha - M^2}}\sinh\left(\phi\sqrt{\dfrac{2m\alpha}{M^2}-1}\right)$

(c) for $E < 0$ and $\dfrac{M^2}{2m} < \alpha$, $\quad \dfrac{1}{r} = \sqrt{\dfrac{2m|E|}{2m\alpha - M^2}}\cosh\left(\phi\sqrt{\dfrac{2m\alpha}{M^2}-1}\right)$

In all three cases

$$t = \frac{1}{E}\sqrt{\frac{m}{2}\left(Er^2 + \alpha - \frac{M^2}{2m}\right)}$$

Problem 9.

When a small correction $\delta U(r)$ is added to the potential energy $U = -\alpha/r$, the paths of finite motion are no longer closed, and at each revolution the perihelion is displaced through a small angle $\delta\phi$. Find $\delta\phi$ when

(a)

$$\delta U = \frac{\beta}{r^2}$$

Solution: From the equation (14.10) of the book, we have

$$\begin{aligned}
\Delta\phi &= 2\int_{r_{\min}}^{r_{\max}} \frac{M dr}{r^2\sqrt{2m(E-U) - \frac{M^2}{r^2}}} \\
&= -2\frac{\partial}{\partial M}\int_{r_{\min}}^{r_{\max}} \sqrt{2m(E-U) - \frac{M^2}{r^2}}\, dr \\
&= -2\frac{\partial}{\partial M}\int_{r_{\min}}^{r_{\max}} \sqrt{2m\left(E + \frac{\alpha}{r} - \delta U\right) - \frac{M^2}{r^2}}\, dr.
\end{aligned}$$

Expanding the integrand in powers of δU involves using a Taylor series expansion. Let's denote the integrand as F:

$$F = \sqrt{2m\left(E + \frac{\alpha}{r} + \delta U\right) - \frac{M^2}{r^2}}$$

We want to expand F around $\delta U = 0$. The expansion will look like:

$$F = F_0 + F_1\delta U + F_2(\delta U)^2 + \dots$$

Here, F_0 is the value of F at $\delta U = 0$, F_1 is the first derivative with respect to δU at $\delta U = 0$, and so on. Let's find the derivatives:

$$\begin{aligned}
F_0 &= \sqrt{2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}} \\
F_1 &= \frac{\partial F}{\partial(\delta U)}\Big|_{\delta U=0} = \frac{m}{\sqrt{2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}}} \\
F_2 &= \frac{\partial^2 F}{\partial(\delta U)^2}\Big|_{\delta U=0} = -\frac{m^2\delta U}{\left(2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}\right)^{3/2}}.
\end{aligned}$$

Therefore, the expanded expression in powers of δU is:

$$F = F_0 + \frac{m}{\sqrt{2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}}}\delta U - \frac{m^2}{2\left(2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}\right)^{3/2}}\delta U^2 + \dots$$

After plugging the expended expression of F in the integral, we see that the zero-order term gives 2π. The first-order term gives the required change $\delta\phi$:

$$\delta\phi = \frac{\partial}{\partial M}\int_{r_{\min}}^{r_{\max}} \frac{2m\delta U}{\sqrt{2m\left(E + \frac{\alpha}{r}\right) - \frac{M^2}{r^2}}}. \tag{3.11}$$

We can change the integration over r to one over ϕ, along the path of the unperturbed motion, using the substitution

$$\phi = \cos^{-1}\frac{(M/r) - (m\alpha/M)}{\sqrt{2mE + \frac{m^2\alpha^2}{M^2}}}$$

with the differential form

$$\begin{aligned}\frac{r^2}{M}d\phi &= \frac{-1}{\sqrt{1-\left(\frac{(M/r)-(m\alpha/M)}{\sqrt{2mE+\frac{m^2\alpha^2}{M^2}}}\right)^2}}\frac{1}{\sqrt{2mE+\frac{m^2\alpha^2}{M^2}}}\\ &= \sqrt{2m\left(E+\frac{\alpha}{r}\right)-\frac{M^2}{r^2}}.\end{aligned}$$

Hence, the integral Eq. (3.11) become

$$\begin{aligned}\delta\phi &= \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi r^2\delta U d\phi\right) \\ &= \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi r^2\frac{\beta}{r^2}d\phi\right)\\ &= \frac{\partial}{\partial M}\left(\frac{2\pi m\beta}{M}\right)\\ &= -\frac{2\pi m\beta}{M^2}.\end{aligned} \tag{3.12}$$

Using the formulae (15.4) of the book we can finally write

Answer to Problem 9 (a).

$$\delta\phi = -\frac{2\pi\beta}{\alpha p}$$

(b)

$$\delta U = \frac{\gamma}{r^3}$$

Solution: From Eq. (3.12), we have

$$\delta\phi = \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi \frac{\gamma}{r}d\phi\right).$$

Using the formulae (15.5) from the book, the integral become

$$\begin{aligned}\delta\phi &= \frac{\partial}{\partial M}\left(\frac{2m}{M}\int_0^\pi \frac{\gamma}{p}\left(1+e\cos\phi\right)d\phi\right)\\ &= \frac{\partial}{\partial M}\left(\frac{2\pi\gamma m}{Mp}\right)\\ &= -\frac{6\pi\alpha\gamma m^2}{M^4}.\end{aligned}$$

The last expression can also be written as

Answer to Problem 9 (b).

$$\delta\phi = -\frac{6\pi\gamma}{\alpha p^2}$$

Chapter 4.

Collisions Between Particles

Problem 1.

Find the relation between the angles θ_1, θ_2 (in the L system) after disintegrating into two particles.

Solution: The angles of the particles, in the C system, are related by

$$\theta_0 = \theta_{10} = \pi - \theta_{20},$$

where θ_0 as been defined to simplify the notation. From the formulae (16.5) of the book, we have

$$\begin{aligned} V + v_{10}\cos\theta_0 &= v_{10}\sin\theta_0\cot\theta_1 \\ V - v_{20}\cos\theta_0 &= v_{20}\sin\theta_0\cot\theta_2. \end{aligned}$$

We must eliminate θ_0 from the two equations above. To do so, we can first solve for $\cos\theta_0$ and $\sin\theta_0$ which give us

$$\sin\theta_0 = \frac{V + v_{10}\cos\theta_0}{v_{10}\cot\theta_1} = \frac{V - v_{20}\cos\theta_0}{v_{20}\cot\theta_2} \tag{4.1}$$

$$\Rightarrow V v_{20}\cot\theta_2 + v_{10}v_{20}\cos\theta_0\cot\theta_2 = V v_{10}\cot\theta_1 - v_{10}v_{20}\cos\theta_0\cot\theta_1$$

$$\Rightarrow \cos\theta_0 = \frac{V\left(v_{10}\cot\theta_1 - v_{20}\cot\theta_2\right)}{v_{10}v_{20}\left(\cot\theta_1 + \cot\theta_2\right)}. \tag{4.2}$$

Using the sum of square of Eq. (4.1) and Eq. (4.2), we get

$$\left(\frac{V + v_{10}\cos\theta_0}{v_{10}\cot\theta_1}\right)^2 + \left(\frac{V\left(v_{10}\cot\theta_1 - v_{20}\cot\theta_2\right)}{v_{10}v_{20}\left(\cot\theta_1 + \cot\theta_2\right)}\right)^2 = 1$$

$$\Rightarrow \left(v_{10} + v_{20}\right)^2 + \left(v_{10}\cot\theta_1 - v_{20}\cot\theta_2\right)^2 = \frac{v_{10}^2 v_{20}^2}{V^2}\left(\cot\theta_1 + \cot\theta_2\right)^2$$

$$\Rightarrow v_{10}^2\csc\theta_1{}^2 + v_{20}^2\csc\theta_2{}^2 - 2v_{10}v_{20}\cot\theta_1\cot\theta_2 = \frac{v_{10}^2 v_{20}^2\sin^2(\theta_1+\theta_2)}{V^2\sin^2\theta_1\sin^2\theta_2}$$

$$\Rightarrow v_{20}^2\sin^2\theta_1 + v_{10}^2\sin^2\theta_2 - 2v_{10}v_{20}\sin\theta_1\sin\theta_2\cos\theta_1\cos\theta_2 = \frac{v_{10}^2 v_{20}^2}{V^2}\sin^2(\theta_1+\theta_2).$$

Using the equation (16.2) from the book and the relation $v_{10}/v_{20} = m_2/m_1$, we obtain

Answer to Problem 1.

$$\frac{m_2}{m_1}\sin^2\theta_2 + \frac{m_1}{m_2}\sin^2\theta_1 - 2\sin\theta_1\sin\theta_2\cos\theta_1\cos\theta_2 = \frac{2\epsilon}{(m_1+m_2)V^2}\sin^2(\theta_1+\theta_2).$$

Problem 2.

Find the angular distribution of the resulting particles in the L system.

Solution: The distribution of the resulting particles with respect to the angle θ_0 is given by the equation (16.7) of the book, *i.e.*

$$\frac{1}{2}\sin\theta_0 d\theta_0 = \frac{1}{2}\mathrm{d}(\cos\theta_0).$$

Using equation (16.6) of the book yield

$$\frac{1}{2}\mathrm{d}(\cos\theta_0) = -\frac{V}{v_0}\sin\theta\cos\theta\mathrm{d}\theta \pm \left(\sin\theta\sqrt{1-(V^2/v_0^2)\sin^2\theta} + \cos\theta\frac{(V^2/v_0^2)\sin\theta\cos\theta}{\sqrt{1-(V^2/v_0^2)\sin^2\theta}}\right)\mathrm{d}\theta.$$

After simplification, the angular distribution become

Answer to Problem 2.

$$\frac{1}{2}\sin\theta\mathrm{d}\theta\left(2\frac{V}{v_0}\cos\theta \pm \frac{1+(V^2/v_0^2)\cos 2\theta}{\sqrt{1-(V^2/v_0^2)\sin^2\theta}}\right)$$

Problem 3.

Determine the range of possible values of the angle θ between the directions of motion of the two resulting particles in the L system.

Solution: The tangent of the angle $\theta = \theta_1 + \theta_2$ is

$$\tan\theta = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 \tan\theta_2}.$$

Using equation (16.5) of the book to get $\tan\theta_1$ and $\tan\theta_1$, and using $\theta_0 = \theta_{10} = \pi - \theta_{20}$, the last equation become

$$\tan\theta = \frac{(v_{10} + v_{20})V \sin\theta_0}{V^2 + (v_{10} - v_{20})V\cos\theta_0 - v_{10}v_{20}}$$

or equivalently

$$\cot\theta = \frac{V^2 + (v_{10} - v_{20})V\cos\theta_0 - v_{10}v_{20}}{(v_{10} + v_{20})V\sin\theta_0}.$$

In order to find the extrema, we take the derivative with respect to θ_0,

$$\frac{\partial\cot\theta}{\partial\theta_0} = 0 = \frac{(v_{20} - v_{10})V^2 - V\cos\theta_0(V^2 - v_{10}v_{20})}{(v_{10} + v_{20})V^2\sin^2\theta_0}$$
$$\Rightarrow \cos\theta_0 = \frac{(v_{20} - v_{10})V}{V^2 - v_{10}v_{20}}.$$

We know that $-1 \leq \cos\theta_0 \leq 1$, thus $v_{10} < V < v_{20}$. Moreover, we could calculate the derivative of the cotangent with respect to V and find the expression:

$$\frac{\partial\cot\theta}{\partial V} = \frac{V^2 + v_{10}v_{20}}{(v_{10} + v_{20})V^2\sin\theta_0}$$

which is positive for all values of V.

This is why $\cot\theta$ may assume all the values between 0 and π when $v_{10} < V < v_{20}$, because $\frac{\partial\cot\theta}{\partial\theta_0} \neq 0$ and $\frac{\partial\cot\theta}{\partial V} \neq 0$ for all V, θ_0 in that interval. It is easy to see that if V is kept constant and θ_0 is varied between 0 and π, $\cot\theta$ spans the interval $(-\infty, +\infty)$; similarly, if θ_0 is kept constant and V is varied between 0 and $+\infty$, $\cot\theta$ spans the interval $(-\infty, +\infty)$.

When $V \leq v_{10}$, it is easy to see that $\cot\theta$ has a maximum value. In fact,

$$\lim_{\theta_0\to 0^+}\cot(x) = -\infty, \quad \lim_{\theta_0\to\pi^-}\cot(x) = -\infty$$

and the maximum value of the cotangent is reached when:

$$\cos\theta_0 = \frac{(v_{20} - v_{10})V}{V^2 - v_{10}v_{20}}$$

The calculation of the maximum value leads to:

$$\cot\theta = \frac{V^2 + (v_{10} - v_{20})V\cos\theta_0 - v_{10}v_{20}}{(v_{10} + v_{20})V\sin\theta_0} = -\frac{\sqrt{(V^2 - v_{10}v_{20})^2 - (v_{20} - v_{10})^2V^2}}{(v_{10} + v_{20})V}$$

Then, after transforming the cotangent into a sine, one gets:

$$\sin\theta = \frac{(v_{20} + v_{10})V}{V^2 + v_{10}v_{20}}$$

If we put:

$$\alpha = \sin^{-1}\left(\frac{(v_{20} + v_{10})V}{V^2 + v_{10}v_{20}}\right)$$

then, in the interval where $V \leq v_{10}$, we have the following ranges for θ: $\pi - \alpha \leq \theta \leq \pi$. Similarly, when $V \geq v_{20}$, it is easy to see that $\cot\theta$ has a minimum value. In fact,

$$\lim_{\theta_0\to 0^+}\cot(x) = +\infty, \quad \lim_{\theta_0\to\pi^-}\cot(x) = +\infty$$

and the minimum value of the cotangent is reached when $\theta = \alpha$. In this case, the ranges for θ are $0 \leq \theta \leq \alpha$ (the equality $\theta = 0$ is satisfied only when $V \to +\infty$). The final solution is thus

Answer to Problem 3.

$$\begin{aligned} &\text{if } v_{10} < V < v_{20} \quad \Rightarrow \quad 0 < \theta < \pi \\ &\text{if } V \leq v_{10} \quad \Rightarrow \quad \pi - \alpha \leq \theta \leq \pi \\ &\text{if } V \geq v_{20} \quad \Rightarrow \quad 0 \leq \theta \leq \alpha \end{aligned}$$

where

$$\alpha = \sin^{-1}\left(\frac{(v_{20} + v_{10})V}{V^2 + v_{10}v_{20}}\right).$$

Problem 4.

Express the velocity of each particle after a collision between a moving particle (m_1) and another at rest (m_2) in terms of their directions of motion in the L system.

Solution: To solve this problem with need to use the Fig. 16 of the book. First, using the law of sines, the magnitude of the momentum of the second particle after the collision is

$$p_2' = \mathrm{OB}\frac{\sin\chi}{\sin\theta_2} = mv\frac{\sin\chi}{\cos(\chi/2)} = 2mv\sin\frac{\chi}{2}.$$

We get the equation (17.5) of the book if we isolate the velocity

$$v_2' = \frac{2m_1 v}{m_1+m_2}\sin\frac{\chi}{2}.$$

Second, using the law of cosines, we get

$$\begin{aligned}
\mathrm{OC}^2 &= \mathrm{AO}^2 + p_1'^2 - 2\mathrm{AO}p_1'\cos\theta_1 \\
\Rightarrow p_1' &= \mathrm{AO}\cos\theta_1 \pm \sqrt{\mathrm{AO}^2\cos^2\theta_1 + \mathrm{OC}^2 - \mathrm{AO}^2} \\
\Rightarrow v_1' &= \frac{m_1\cos\theta_1 \pm \sqrt{m_2^2 - m_1^2\sin^2\theta_1}}{m_1+m_2}v_1.
\end{aligned}$$

For the case where $m_2 > m_1$ we also get the equation (17.5) of the book, but when $m_1 > m_2$, the radical may have either sign. To summarize, the answer is

Answer to Problem 4.

$$\begin{aligned}
v_2' &= \frac{2m_1 v}{m_1+m_2}\sin\frac{\chi}{2} \\
v_1' &= \frac{m_1\cos\theta_1 \pm \sqrt{m_2^2 - m_1^2\sin^2\theta_1}}{m_1+m_2}v_1.
\end{aligned}$$

Problem 5.

Determine the effective cross-section for scattering of particles from a perfectly rigid sphere of radius a (*i.e.* when the interaction is such that $U = \infty$ for $r < a$ and $U = 0$ for $r > a$).

Solution: From Fig. 19 of the book, we can see

$$\rho = a\sin\phi_0 = a\sin\frac{1}{2}(\pi - \chi) = a\cos\frac{1}{2}\chi. \tag{4.3}$$

Subsituing Eq. (4.3) in equation (18.7) or (18.8) of the book, we have

$$\mathrm{d}\sigma = \frac{1}{2}\pi a^2\sin\chi\mathrm{d}\chi = \frac{1}{4}a^2\mathrm{d}o. \tag{4.4}$$

If we integrate $\mathrm{d}\sigma$ over all angles, we find

$$\sigma = \frac{1}{2}\pi a^2\int_{-\pi/2}^{\pi/2}\sin\chi\mathrm{d}\chi = \pi a^2,$$

which is the projection of the sphere perpendicular to the the beam (*i.e.* the impact area). We need to express Eq. (4.4) in the L system using the equation (17.4) of the book. If this equation is solve for $\cos\theta_1$, we get

$$\begin{aligned}
&\tan\theta_1 = \frac{m_2\sin\chi}{m_1 + m_2\cos\chi}\\
&\Rightarrow \tan^2\theta_1\left(\frac{m_1}{m_2} + \cos\chi\right)^2 = 1 - \cos^2\chi\\
&\Rightarrow \cos\chi = -\frac{m_1}{m_2}\sin^2\theta_1 \pm \cos\theta_1\sqrt{1 - \frac{m_1^2}{m_2^2}\sin^2\theta_1}\\
&\Rightarrow \frac{1}{2}\sin\chi\mathrm{d}\chi = \frac{1}{2}\sin\theta_1\mathrm{d}\theta_1\left(2\frac{m_1}{m_2}\cos\theta_1 \pm \frac{1 + (m_1^2/m_2^2)\cos 2\theta_1}{\sqrt{1 - (m_1^2/m_2^2)\sin^2\theta_1}}\right).
\end{aligned}$$

We thus get

Answer to Problem 5.

$$\begin{aligned}
\mathrm{d}\sigma_1 &= \frac{1}{4}a^2\left(2\frac{m_1}{m_2}\cos\theta_1 \pm \frac{1 + (m_1^2/m_2^2)\cos 2\theta_1}{\sqrt{1 - (m_1^2/m_2^2)\sin^2\theta_1}}\right)\mathrm{d}o_1\\
\mathrm{d}\sigma_2 &= a^2|\cos\theta_2|\mathrm{d}o_2.
\end{aligned}$$

Problem 6.

Express the effective cross-section (see last problem) as a function of the energy ϵ lost by a scattered particle.

Solution: The energy loss by a particle of mass m_1 is equal to the energy gained by the sphere of mass m_2. Knowing that the sphere is initially at rest, the energy gained is

$$\epsilon = E_2' = \frac{1}{2}m_2 v_2'^2 = \frac{2m_1^2 m_2}{(m_1+m_2)^2} v_\infty^2 \sin\frac{1}{2}\chi = \epsilon_{max}\sin\frac{1}{2}\chi.$$

Using the double angle identity, the variation of ϵ is

$$\mathrm{d}\epsilon = \epsilon_{max}\sin\frac{1}{2}\chi\cos\frac{1}{2}\chi\mathrm{d}\chi = \frac{1}{2}\epsilon_{max}\sin\chi\mathrm{d}\chi.$$

Using Eq. (4.4) from the last problem, the effective cross section can thus be express as

Answer to Problem 6.

$$\mathrm{d}\sigma = \frac{\pi a^2 \mathrm{d}\epsilon}{\epsilon_{max}}$$

Problem 7.

Find the effective cross-section as a function of the velocity v_∞ for particles scattered in a field $U \sim r^{-n}$.

Solution: For a central potential field, the scattering angle is determined by the integral

$$\chi = \pi - 2\int_{r_{min}}^{\infty} \frac{\rho \mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}}.$$

Subsituing $U(r) = \alpha r^{-n}$ and define the dimensionless variable $x = \frac{r}{\rho}$, then

$$\chi = \pi - 2\int_{x_{min}}^{\infty} \frac{\mathrm{d}x}{x^2\sqrt{1-\frac{1}{x^2}-\frac{2\alpha}{mv_\infty^2\rho^n x^n}}}.$$

Notice the dimensionless combination $\xi = \frac{2\alpha}{mv_\infty^2\rho^n}$, the integral simplifies to

$$\chi = \pi - 2\int_{x_{min}}^{\infty} \frac{\mathrm{d}x}{x^2\sqrt{1-\frac{1}{x^2}-\frac{\xi}{x^n}}}.$$

Thus, the scattering angle depends only on ξ,

$$\chi = f(\xi),$$

where $f(\xi)$ is a dimensionless function determined by the integral. To invert this relationship, solve for ρ in terms of χ and v_∞,

$$\xi = \frac{2\alpha}{mv_\infty^2\rho^n} \Rightarrow \rho = \left(\frac{2\alpha}{mv_\infty^2\xi}\right)^{1/n}.$$

Let $\xi = g(\chi)$, where $g(\chi)$ is a function encoding the angular dependence. Then,

$$\rho = \left(\frac{2\alpha}{mv_\infty^2}\right)^{1/n} (g(\chi))^{-1/n}$$
$$\Rightarrow \rho \sim v_\infty^{-2/n} (g(\chi))^{-1/n}.$$

The differential cross-section is thus

$$\mathrm{d}\sigma = \frac{\rho}{\sin\chi}\left|\frac{\mathrm{d}\rho}{\mathrm{d}\chi}\right|\mathrm{d}o$$
$$\mathrm{d}\sigma \sim v_\infty^{-4/n}\frac{F(\chi)}{\sin\chi}\mathrm{d}o,$$

where $F(\chi) = \left|\frac{\mathrm{d}}{\mathrm{d}\chi}(g(\chi))^{-1/n}\right|$. Integrate over the solid angle

$$\sigma \sim v_\infty^{-4/n}\int F(\chi)\mathrm{d}\chi.$$

Knowing that the integral depends only on n, not v_∞, we have

Answer to Problem 7.

$$\sigma \sim v_\infty^{-4/n}$$

Problem 8.

Determine the effective cross-section for a particle to "fall" to the centre of a field $U = -\alpha/r^2$.

Solution: Let the particle come from infinity with velocity v_∞ and impact parameter ρ, and take $\alpha > 0$. Its energy and angular momentum are

$$E = \frac{1}{2}mv_\infty^2, \qquad M = mv_\infty\rho.$$

The radial motion is governed by the energy equation

$$\frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r) = E,$$

where the effective potential energy is

$$U_{\text{eff}}(r) = \frac{M^2}{2mr^2} - \frac{\alpha}{r^2} = \frac{1}{r^2}\left(\frac{M^2}{2m} - \alpha\right).$$

Since both terms vary as r^{-2}, the sign of the bracket determines the behaviour of the particle near the centre. If $M^2 > 2m\alpha$, then U_{eff} is a repulsive centrifugal barrier which tends to $+\infty$ as $r \to 0$, so the particle reaches a turning point where $U_{\text{eff}}(r_{min}) = E$ and goes back to infinity. If $M^2 < 2m\alpha$, then $U_{\text{eff}} < 0$ for all r, so $E - U_{\text{eff}} > 0$ everywhere. The radial velocity never vanishes and an incoming particle has no radial turning point: it falls to the centre. In the limiting case $M^2 = 2m\alpha$, we have $U_{\text{eff}} = 0$, thus $\dot{r} = -v_\infty$ and the particle also reaches the centre. The critical trajectory falls to the centre because the centrifugal and attractive terms have the same dependence r^{-2}. At the critical value of M they cancel identically, so there is no barrier at all and nothing can stop the particle at a finite distance. This differs from the next problem, where the two terms have different powers of r and U_{eff} has a maximum at a finite radius.

The particle therefore falls to the centre if $M^2 \leq 2m\alpha$, *i.e.* if

$$m^2v_\infty^2\rho^2 \leq 2m\alpha \Rightarrow \rho \leq \rho_{max}, \qquad \rho_{max}^2 = \frac{2\alpha}{mv_\infty^2}.$$

The effective cross-section is obtained by summing the areas of all the annuli of the incident beam that lead to a fall,

$$\sigma = \int_0^{\rho_{max}} 2\pi\rho\,\mathrm{d}\rho = \pi\rho_{max}^2.$$

We thus get

Answer to Problem 8.

$$\sigma = \frac{2\pi\alpha}{mv_\infty^2} = \frac{\pi\alpha}{E}$$

Problem 9.

The same as last problem, but for a field $U = -\alpha/r^n$ $(n > 2, \alpha > 0)$.

Solution: As in the last problem, the particle comes from infinity with velocity v_∞ and impact parameter ρ, so that

$$E = \frac{1}{2}mv_\infty^2, \qquad M = mv_\infty\rho.$$

The effective potential energy is now

$$U_{\text{eff}}(r) = \frac{M^2}{2mr^2} - \frac{\alpha}{r^n}.$$

Since $n > 2$, the centrifugal term dominates at large r and the attractive term dominates at small r. Thus $U_{\text{eff}} \to 0^+$ as $r \to \infty$, $U_{\text{eff}} \to -\infty$ as $r \to 0$, and U_{eff} has a maximum at some r_b, given by

$$U'_{\text{eff}}(r_b) = -\frac{M^2}{mr_b^3} + \frac{n\alpha}{r_b^{n+1}} = 0 \Rightarrow \frac{M^2}{m} = \frac{n\alpha}{r_b^{n-2}}.$$

This barrier analysis assumes $M \neq 0$. A head-on particle ($\rho = 0$) has no centrifugal barrier: $U_{\text{eff}} = -\alpha/r^n$ is monotonic, and the particle falls to the centre. The height of this barrier is

$$U_{\text{eff}}(r_b) = \frac{n\alpha}{2r_b^n} - \frac{\alpha}{r_b^n} = \frac{(n-2)\alpha}{2r_b^n}.$$

If E is smaller than the barrier height, the particle reaches a turning point outside the barrier and goes back to infinity. If E exceeds the barrier height, then $E - U_{\text{eff}} > 0$ for all r, so the particle has no radial turning point and falls to the centre. Since the barrier height increases with M, the capture occurs for $\rho < \rho_{max}$, where ρ_{max} is the impact parameter for which E is equal to the barrier height. In this limiting case, the incoming particle approaches asymptotically the unstable circular orbit $r = r_b$. Indeed, at $r = r_b$ both $E - U_{\text{eff}}$ and U'_{eff} vanish, so r_b is a point of (unstable) equilibrium of the radial motion. Near it, $\frac{1}{2}m\dot{r}^2 = E - U_{\text{eff}} \propto (r - r_b)^2$, so the time $\int \mathrm{d}r/|\dot{r}| \propto \int \mathrm{d}r/|r - r_b|$ needed to reach r_b diverges logarithmically. In contrast with the last problem, the critical particle therefore never reaches the centre. This single value of ρ does not affect the cross-section.

Denote by r_c the position of the maximum at the critical impact parameter. The condition $E = U_{\text{eff}}(r_c)$ gives

$$\frac{1}{2}mv_\infty^2 = \frac{(n-2)\alpha}{2r_c^n} \Rightarrow r_c^n = \frac{(n-2)\alpha}{mv_\infty^2}.$$

Using $M = mv_\infty\rho_{max}$ in the condition for the maximum, we find

$$\begin{aligned}\rho_{max}^2 &= \frac{M^2}{m^2v_\infty^2} = \frac{n\alpha}{mv_\infty^2 r_c^{n-2}} \\ &= \frac{n}{n-2}\frac{r_c^n}{r_c^{n-2}} = \frac{n}{n-2}r_c^2,\end{aligned}$$

where we used $\alpha/(mv_\infty^2) = r_c^n/(n-2)$ in the second line. The effective cross-section is obtained by summing the areas of all the annuli of the incident beam that lead to a fall,

$$\sigma = \int_0^{\rho_{max}} 2\pi\rho\,\mathrm{d}\rho = \pi\rho_{max}^2 = \frac{\pi n}{n-2}r_c^2.$$

We thus get

Answer to Problem 9.

$$\sigma = \frac{\pi n}{n-2}\left(\frac{(n-2)\alpha}{mv_\infty^2}\right)^{2/n} = \frac{\pi n}{n-2}\left(\frac{(n-2)\alpha}{2E}\right)^{2/n}$$

Problem 10.

Determine the effective cross-section for particles of mass m_1 to strike a sphere of mass m_2 and radius R to which they are attracted in accordance with Newton's law.

Solution: We consider the relative motion of the particle and the sphere, with r the distance between their centres. This is equivalent to the motion of a single particle of reduced mass

$$m = \frac{m_1 m_2}{m_1 + m_2}$$

in the field

$$U(r) = -\frac{\gamma m_1 m_2}{r},$$

where γ is the gravitational constant. Let v_∞ be the relative velocity at infinity and ρ the impact parameter. The energy and angular momentum of the relative motion are

$$E = \frac{1}{2} m v_\infty^2, \qquad M = m v_\infty \rho,$$

and the radial motion is governed by the energy equation

$$\frac{1}{2} m \dot{r}^2 + \frac{M^2}{2mr^2} - \frac{\gamma m_1 m_2}{r} = E.$$

The distance of closest approach decreases as ρ decreases, so the particle strikes the sphere if $\rho \leq \rho_{max}$, where ρ_{max} corresponds to the limiting trajectory which grazes the sphere. For this trajectory, $\dot{r} = 0$ at $r = R$. Substituting $M = m v_\infty \rho_{max}$, we have

$$\frac{m v_\infty^2 \rho_{max}^2}{2R^2} - \frac{\gamma m_1 m_2}{R} = \frac{1}{2} m v_\infty^2.$$

Multiplying by $2R^2/(m v_\infty^2)$ and using $m_1 m_2 / m = m_1 + m_2$,

$$\begin{aligned} \rho_{max}^2 &= R^2 \left(1 + \frac{2\gamma m_1 m_2}{m R v_\infty^2} \right) \\ &= R^2 \left(1 + \frac{2\gamma (m_1 + m_2)}{R v_\infty^2} \right). \end{aligned}$$

The effective cross-section is obtained by summing the areas of all the annuli of the incident beam that lead to a collision with the sphere,

$$\sigma = \int_0^{\rho_{max}} 2\pi \rho \, \mathrm{d}\rho = \pi \rho_{max}^2 = \pi R^2 \left(1 + \frac{2\gamma (m_1 + m_2)}{R v_\infty^2} \right).$$

The second term describes the gravitational focusing of the trajectories towards the sphere. In the high-speed limit $v_\infty^2 \gg \gamma (m_1 + m_2)/R$, it becomes negligible and we recover the geometric cross-section $\sigma = \pi R^2$. As in the book, we assume $m_2 \gg m_1$, so that $m_1 + m_2 \simeq m_2$. We thus get

Answer to Problem 10.

$$\sigma \simeq \pi R^2 \left(1 + \frac{2\gamma m_2}{R v_\infty^2} \right)$$

Problem 11.

Deduce the form of a scattering field $U(r)$, given the effective cross-section as a function of the angle of scattering for a given energy E. It is assumed that $U(r)$ decreases monotonically with r (a repulsive field), with $U(0) > E$ and $U(\infty) = 0$.

Solution: Let $\mathrm{d}\sigma/\mathrm{d}\theta$ be the effective cross-section per unit scattering angle, *i.e.* already integrated over the azimuth. In terms of the cross-section per unit solid angle, since $\mathrm{d}o = 2\pi \sin\theta \, \mathrm{d}\theta$,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\theta} = 2\pi \sin\theta \, \frac{\mathrm{d}\sigma}{\mathrm{d}o}.$$

As in the book, we assume that the relation between ρ and χ is one-to-one, with χ decreasing monotonically from π (head-on collision, $\rho = 0$) to 0 (no deflection, $\rho = \infty$). Note that this is an additional assumption: the monotonicity of U alone does not guarantee it. The particles scattered through angles between χ and π are then exactly those with impact parameters smaller than $\rho(\chi)$, so they fill a disc of area $\pi\rho^2$ in the incident beam, and

$$\pi\rho^2(\chi) = \int_\chi^\pi \frac{\mathrm{d}\sigma}{\mathrm{d}\theta} \, \mathrm{d}\theta.$$

Knowing the cross-section, this integral gives $\rho(\chi)$, which we invert to get the scattering angle $\chi(\rho)$ as a function of the impact parameter. The problem is thus reduced to finding $U(r)$ from $\chi(\rho)$.

The scattering angle in a central field is given by

$$\chi = \pi - 2\int_{r_{min}}^\infty \frac{\rho \, \mathrm{d}r}{r^2\sqrt{1 - \frac{\rho^2}{r^2} - \frac{U(r)}{E}}}.$$

We now introduce the variables

$$s = \frac{1}{r}, \qquad x = \frac{1}{\rho^2}, \qquad w(s) = \sqrt{1 - \frac{U(1/s)}{E}},$$

and write $\widehat{\chi}(x) = \chi(x^{-1/2})$ for the same scattering angle expressed in terms of x. Since $\mathrm{d}r = -\mathrm{d}s/s^2$, we have $\rho \, \mathrm{d}r/r^2 = -\rho \, \mathrm{d}s$. When r goes from r_{min} to ∞, s goes from $s_0 = 1/r_{min}$ to 0, and the minus sign is absorbed by reversing the limits. Thus

$$\frac{\pi - \widehat{\chi}(x)}{2} = \int_0^{s_0} \frac{\rho \, \mathrm{d}s}{\sqrt{w^2(s) - \rho^2 s^2}} = \int_0^{s_0(x)} \frac{\mathrm{d}s}{\sqrt{x w^2(s) - s^2}},$$

where we divided the numerator and the denominator by $\rho = x^{-1/2}$. The turning point $s_0(x)$ is the zero of the radicand,

$$x w^2(s_0) - s_0^2 = 0.$$

Since U decreases with r, it increases with s, so $w(s)$ decreases and $s/w(s)$ increases with s. The radicand is therefore positive for $0 \le s < s_0(x)$, and the condition $s < s_0(x)$ is equivalent to $x > s^2/w^2(s)$.

This is an integral equation of Abel's type, which we solve as follows. Let A be an auxiliary parameter (denoted α in the book). We divide both sides by $\sqrt{A - x}$ and integrate over x from 0 to A,

$$\frac{1}{2}\int_0^A \frac{\pi - \widehat{\chi}(x)}{\sqrt{A - x}} \, \mathrm{d}x = \int_0^A \int_0^{s_0(x)} \frac{\mathrm{d}s \, \mathrm{d}x}{\sqrt{(x w^2(s) - s^2)\,(A - x)}}.$$

The domain of integration on the right is $0 < x < A$, $0 < s < s_0(x)$. From the previous remark, it can equally be described by $0 < s < s_0(A)$ and $s^2/w^2(s) < x < A$. Reversing the order of integration and writing $x w^2 - s^2 = w^2(x - s^2/w^2)$, we get

$$\frac{1}{2}\int_0^A \frac{\pi - \widehat{\chi}(x)}{\sqrt{A - x}} \, \mathrm{d}x = \int_0^{s_0(A)} \frac{\mathrm{d}s}{w(s)} \int_{s^2/w^2(s)}^A \frac{\mathrm{d}x}{\sqrt{(x - s^2/w^2(s))\,(A - x)}}.$$

For the inner integral, let $a = s^2/w^2$ and substitute $x = a + (A - a)\sin^2\varphi$, so that $\mathrm{d}x = 2(A - a)\sin\varphi\cos\varphi \, \mathrm{d}\varphi$ and $\sqrt{(x - a)(A - x)} = (A - a)\sin\varphi\cos\varphi$. Then

$$\int_a^A \frac{\mathrm{d}x}{\sqrt{(x - a)(A - x)}} = \int_0^{\pi/2} 2 \, \mathrm{d}\varphi = \pi,$$

independently of a and A. Hence

$$\frac{1}{2}\int_0^A \frac{\pi-\widehat{\chi}(x)}{\sqrt{A-x}}\,\mathrm{d}x = \pi\int_0^{s_0(A)}\frac{\mathrm{d}s}{w(s)}.$$

On the left, the first term gives $\frac{1}{2}\pi\int_0^A (A-x)^{-1/2}\,\mathrm{d}x = \pi\sqrt{A}$. For the second term, we integrate by parts with $\mathrm{d}x/\sqrt{A-x} = -\mathrm{d}\left(2\sqrt{A-x}\right)$,

$$\begin{aligned}\int_0^A \frac{\widehat{\chi}(x)\,\mathrm{d}x}{\sqrt{A-x}} &= \left[-2\widehat{\chi}(x)\sqrt{A-x}\right]_0^A + 2\int_0^A \widehat{\chi}'(x)\sqrt{A-x}\,\mathrm{d}x \\ &= 2\int_0^A \widehat{\chi}'(x)\sqrt{A-x}\,\mathrm{d}x,\end{aligned}$$

since the boundary term vanishes at $x=A$ and $\widehat{\chi}(0)=0$ (a particle with infinite impact parameter is not deflected). Therefore

$$\pi\sqrt{A} - \int_0^A \widehat{\chi}'(x)\sqrt{A-x}\,\mathrm{d}x = \pi\int_0^{s_0(A)}\frac{\mathrm{d}s}{w(s)}.$$

We differentiate this with respect to A. The integrand on the left vanishes at the upper limit $x=A$, so only the derivative of $\sqrt{A-x}$ contributes,

$$\frac{\pi}{2\sqrt{A}} - \frac{1}{2}\int_0^A \frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{A-x}} = \frac{\pi}{w(s_0)}\frac{\mathrm{d}s_0}{\mathrm{d}A}.$$

This holds for every A. Since A and s_0 are related by $Aw^2(s_0) = s_0^2$, we may instead take s_0 as the independent variable, rename it s, and put $A = s^2/w^2$ with $w = w(s)$. Multiplying by $\mathrm{d}A = 2(s/w)\,\mathrm{d}(s/w)$ and using $\sqrt{A} = s/w$, we get

$$\pi\,\mathrm{d}\left(\frac{s}{w}\right) - \frac{s}{w}\,\mathrm{d}\left(\frac{s}{w}\right)\int_0^{s^2/w^2}\frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{s^2/w^2-x}} = \frac{\pi\,\mathrm{d}s}{w}.$$

Since

$$\pi\,\mathrm{d}\left(\frac{s}{w}\right) - \frac{\pi\,\mathrm{d}s}{w} = -\frac{\pi s\,\mathrm{d}w}{w^2} = \frac{s}{w}\left(-\pi\,\mathrm{d}\log w\right),$$

dividing by s/w gives

$$-\pi\,\mathrm{d}\log w = \mathrm{d}\left(\frac{s}{w}\right)\int_0^{s^2/w^2}\frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{s^2/w^2-x}}.$$

Let $h = s/w$, which increases from 0 with s. At $s=0$ (*i.e.* $r=\infty$), $U=0$ and $w=1$, so $\log w = 0$. Integrating from $s=0$, and calling η the variable of integration for h,

$$-\pi\log w = \int_0^h \mathrm{d}\eta\int_0^{\eta^2}\frac{\widehat{\chi}'(x)\,\mathrm{d}x}{\sqrt{\eta^2-x}}.$$

The domain $0<\eta<h$, $0<x<\eta^2$ is also $0<x<h^2$, $\sqrt{x}<\eta<h$. Interchanging the integrations,

$$-\pi\log w = \int_0^{h^2}\widehat{\chi}'(x)\left[\int_{\sqrt{x}}^h\frac{\mathrm{d}\eta}{\sqrt{\eta^2-x}}\right]\mathrm{d}x = \int_0^{h^2}\widehat{\chi}'(x)\,\operatorname{arcosh}\left(\frac{h}{\sqrt{x}}\right)\mathrm{d}x,$$

where we used $\int \mathrm{d}\eta/\sqrt{\eta^2-x} = \operatorname{arcosh}(\eta/\sqrt{x})$ and $\operatorname{arcosh}1 = 0$.

We now go back to the variables r and ρ. We have $h = 1/(rw)$ and $\sqrt{x} = 1/\rho$, so $h/\sqrt{x} = \rho/(rw)$, and $\widehat{\chi}'(x)\,\mathrm{d}x = \mathrm{d}\chi = (\mathrm{d}\chi/\mathrm{d}\rho)\,\mathrm{d}\rho$. The limit $x=0$ corresponds to $\rho=\infty$ and $x=h^2$ to $\rho = rw$. Hence

$$-\pi\log w = \int_\infty^{rw}\operatorname{arcosh}\left(\frac{\rho}{rw}\right)\frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho = -\int_{rw}^\infty \operatorname{arcosh}\left(\frac{\rho}{rw}\right)\frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho,$$

i.e.

$$w(r) = \exp\left\{\frac{1}{\pi}\int_{rw(r)}^\infty \operatorname{arcosh}\left(\frac{\rho}{rw(r)}\right)\frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho\right\}.$$

With the assumption adopted at the start, that χ decreases monotonically with ρ, we have $\mathrm{d}\chi/\mathrm{d}\rho < 0$. The exponent is then negative and $w < 1$, *i.e.* $U > 0$, as expected for a repulsive field.

To obtain the second form, we integrate by parts in ρ, keeping r and $w(r)$ fixed. Since

$$\frac{\mathrm{d}}{\mathrm{d}\rho}\operatorname{arcosh}\left(\frac{\rho}{rw}\right) = \frac{1}{\sqrt{\rho^2 - r^2w^2}},$$

we have

$$\int_{rw}^{\infty} \operatorname{arcosh}\left(\frac{\rho}{rw}\right)\frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho = \left[\chi(\rho)\operatorname{arcosh}\left(\frac{\rho}{rw}\right)\right]_{rw}^{\infty} - \int_{rw}^{\infty}\frac{\chi(\rho)\,\mathrm{d}\rho}{\sqrt{\rho^2 - r^2w^2}}.$$

Two conditions on the behaviour of $\chi(\rho)$ at large ρ are needed here. **(a)** The boundary term vanishes at $\rho = rw$ because $\operatorname{arcosh} 1 = 0$. At $\rho \to \infty$, $\operatorname{arcosh}(\rho/rw) \simeq \log(2\rho/rw)$ grows logarithmically, so the boundary term vanishes there only if $\chi(\rho)\log\rho \to 0$. **(b)** The remaining integral must converge. Its integrand behaves as $\chi(\rho)/\rho$ at large ρ, so at infinity its convergence is equivalent to that of $\int^{\infty}\chi(\rho)\,\mathrm{d}\rho/\rho$. Near the lower limit, the singularity $1/\sqrt{\rho - rw}$ is integrable. Condition (a) alone does not guarantee (b): for example, $\chi = 1/(\log\rho\log\log\rho)$ satisfies $\chi\log\rho \to 0$, but $\int^{\infty}\chi\,\mathrm{d}\rho/\rho$ diverges. A sufficient condition for both is $\chi(\rho) = O(\rho^{-\varepsilon})$ with some $\varepsilon > 0$: then $\chi\log\rho = O(\rho^{-\varepsilon}\log\rho) \to 0$, and $\int^{\infty}\rho^{-1-\varepsilon}\,\mathrm{d}\rho$ converges. For instance, for a field decreasing as r^{-n} at large distances, $\chi \propto \rho^{-n}$ for large ρ (see Problem 15). Under these conditions,

$$w(r) = \exp\left\{-\frac{1}{\pi}\int_{rw(r)}^{\infty}\frac{\chi(\rho)\,\mathrm{d}\rho}{\sqrt{\rho^2 - r^2w^2(r)}}\right\}.$$

Both formulas give w only implicitly, since w also appears in the lower limit. However, the right-hand sides depend on r and w only through the product rw. They therefore give w as an explicit function of rw, from which $w(r)$ follows, and then

$$U(r) = E\left[1 - w^2(r)\right].$$

The derivation covered all s between 0 and the turning points s_0. The smallest distance reached is that of the head-on collision ($\rho = 0$), r_{min}, for which $U(r_{min}) = E$ and $w = 0$. The potential is thus determined in the accessible region $r > r_{min}$. The scattering data give no information on the interior region $r < r_{min}$, where $U > E$. In summary, we get

Answer to Problem 11.

$$w(r) = \exp\left\{\frac{1}{\pi}\int_{rw(r)}^{\infty}\operatorname{arcosh}\left(\frac{\rho}{rw(r)}\right)\frac{\mathrm{d}\chi}{\mathrm{d}\rho}\,\mathrm{d}\rho\right\} = \exp\left\{-\frac{1}{\pi}\int_{rw(r)}^{\infty}\frac{\chi(\rho)\,\mathrm{d}\rho}{\sqrt{\rho^2 - r^2w^2(r)}}\right\},$$

$$w(r) = \sqrt{1 - \frac{U(r)}{E}}, \qquad U(r) = E\left[1 - w^2(r)\right], \qquad r > r_{min}$$

Problem 12.

Find the effective cross-section for scattering in a field $U = \alpha/r^2$ ($\alpha > 0$).

Solution: Let the particle of mass m come from infinity with velocity v_∞ and impact parameter ρ. It is convenient to introduce the positive length a defined by

$$a^2 = \frac{2\alpha}{mv_\infty^2}.$$

With $U(r) = \alpha/r^2$, we have $2U/(mv_\infty^2) = a^2/r^2$, so the angle between the direction of closest approach and the asymptote of the trajectory is

$$\phi_0 = \int_{r_{min}}^{\infty} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1 - \frac{\rho^2}{r^2} - \frac{a^2}{r^2}}} = \int_{r_{min}}^{\infty} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1 - (\rho^2 + a^2)/r^2}}.$$

The distance of closest approach is the zero of the radicand,

$$r_{min} = \sqrt{\rho^2 + a^2}.$$

We substitute $u = 1/r$, so that $\mathrm{d}r/r^2 = -\mathrm{d}u$. When r goes from r_{min} to ∞, u goes from $1/\sqrt{\rho^2 + a^2}$ to 0, and the minus sign is absorbed by reversing the limits,

$$\begin{aligned}\phi_0 &= \rho\int_0^{1/\sqrt{\rho^2+a^2}} \frac{\mathrm{d}u}{\sqrt{1 - (\rho^2 + a^2)u^2}} \\ &= \frac{\rho}{\sqrt{\rho^2 + a^2}}\left[\arcsin\left(\sqrt{\rho^2 + a^2}\,u\right)\right]_0^{1/\sqrt{\rho^2+a^2}} = \frac{\pi\rho}{2\sqrt{\rho^2 + a^2}}.\end{aligned}$$

The scattering angle is therefore

$$\chi = \pi - 2\phi_0 = \pi\left(1 - \frac{\rho}{\sqrt{\rho^2 + a^2}}\right) = \pi\left[1 - \left(1 + \frac{2\alpha}{mv_\infty^2\rho^2}\right)^{-1/2}\right].$$

Since $\rho/\sqrt{\rho^2 + a^2}$ increases from 0 to 1 as ρ goes from 0 to ∞, χ decreases monotonically from π for a head-on collision to 0 as $\rho \to \infty$. Thus each angle $0 < \chi < \pi$ corresponds to a single impact parameter, and we can invert this relation. Writing it as

$$\frac{\rho}{\sqrt{\rho^2 + a^2}} = \frac{\pi - \chi}{\pi},$$

and squaring, we get $\pi^2\rho^2 = (\pi - \chi)^2(\rho^2 + a^2)$, *i.e.*

$$\rho^2\left[\pi^2 - (\pi - \chi)^2\right] = a^2(\pi - \chi)^2 \Rightarrow \rho^2 = a^2\frac{(\pi - \chi)^2}{\chi(2\pi - \chi)},$$

where we used $\pi^2 - (\pi - \chi)^2 = \chi(2\pi - \chi)$. To differentiate, let $y = \pi - \chi$, so that $\rho^2 = a^2y^2/(\pi^2 - y^2)$ and

$$\frac{\mathrm{d}(\rho^2)}{\mathrm{d}y} = a^2\frac{2y(\pi^2 - y^2) + 2y^3}{(\pi^2 - y^2)^2} = \frac{2\pi^2a^2y}{(\pi^2 - y^2)^2}.$$

Since $\mathrm{d}y = -\mathrm{d}\chi$,

$$\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi} = -\frac{2\pi^2a^2(\pi - \chi)}{\chi^2(2\pi - \chi)^2}.$$

The particles scattered into the range $\mathrm{d}\chi$ are those passing through an annulus of radius ρ and width $|\mathrm{d}\rho|$ in the incident beam, so

$$\mathrm{d}\sigma = 2\pi\rho\left|\frac{\mathrm{d}\rho}{\mathrm{d}\chi}\right|\mathrm{d}\chi = \pi\left|\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\right|\mathrm{d}\chi.$$

In terms of the element of solid angle $\mathrm{d}o = 2\pi\sin\chi\,\mathrm{d}\chi$, this becomes

$$\mathrm{d}\sigma = \frac{1}{2\sin\chi}\left|\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\right|\mathrm{d}o = \pi^2a^2\frac{\pi - \chi}{\chi^2(2\pi - \chi)^2}\frac{\mathrm{d}o}{\sin\chi}.$$

This is a differential cross-section. Its integral over all angles diverges at $\chi = 0$, since particles with arbitrarily large impact parameters are still deflected, by arbitrarily small angles. As a check, for $\rho \gg a$ we have $\rho/\sqrt{\rho^2 + a^2} \simeq 1 - a^2/(2\rho^2)$, hence

$$\chi \simeq \frac{\pi a^2}{2\rho^2} = \frac{\pi\alpha}{m v_\infty^2 \rho^2}, \qquad \frac{\mathrm{d}\sigma}{\mathrm{d}o} \simeq \pi^2 a^2 \frac{\pi}{4\pi^2\chi^3} = \frac{\pi\alpha}{2m v_\infty^2 \chi^3}.$$

The same small-angle result follows directly from $\rho^2 \simeq \pi\alpha/(m v_\infty^2 \chi)$, which gives $\mathrm{d}\sigma = \pi|\mathrm{d}(\rho^2)/\mathrm{d}\chi|\,\mathrm{d}\chi = \pi^2\alpha\,\mathrm{d}\chi/(m v_\infty^2 \chi^2)$ with $\mathrm{d}o \simeq 2\pi\chi\,\mathrm{d}\chi$. Substituting a^2, we thus get

Answer to Problem 12.

$$\chi = \pi\left[1 - \left(1 + \frac{2\alpha}{m v_\infty^2 \rho^2}\right)^{-1/2}\right],$$

$$\mathrm{d}\sigma = \frac{2\pi^2\alpha}{m v_\infty^2} \frac{\pi - \chi}{\chi^2(2\pi - \chi)^2} \frac{\mathrm{d}o}{\sin\chi}, \qquad 0 < \chi < \pi$$

Problem 13.

Find the effective cross-section for scattering by a spherical "potential well" of radius a and "depth" U_0, *i.e.* in the field

$$U(r) = \begin{cases} -U_0, & r < a, \\ 0, & r > a, \end{cases} \qquad U_0 > 0.$$

Solution: Let m be the mass of the particle and v_∞ its velocity at infinity. Since U is constant outside the sphere, the particle moves with the constant velocity v_∞ until it reaches $r = a$. Inside the well, U is again constant, so the particle moves with a constant velocity v_{in} given by the conservation of energy,

$$\frac{1}{2}mv_{\text{in}}^2 - U_0 = \frac{1}{2}mv_\infty^2.$$

As in the book, we define the ratio

$$n = \frac{v_{\text{in}}}{v_\infty} = \sqrt{1 + \frac{2U_0}{mv_\infty^2}} > 1.$$

The trajectory is therefore made of straight segments, and its direction changes only at the boundary $r = a$, where the particle receives an impulse directed along the radius. This is analogous to the refraction of a light ray by a sphere with refractive index n.

The geometry of the path is shown in Fig. 21 of the book. Let α be the angle of incidence and β the angle of refraction, both measured from the normal to the sphere (*i.e.* the radius) at the point of entry. At $r = a$, the angular momentum is conserved (the force is radial), so the component of the velocity perpendicular to the radius is the same on both sides,

$$mv_\infty a \sin\alpha = mv_{\text{in}} a \sin\beta \Rightarrow \frac{\sin\alpha}{\sin\beta} = n.$$

Particles with $\rho > a$ do not touch the sphere and are not deflected. For $\rho < a$, the impact parameter is the distance from the centre to the incident line, so

$$\rho = a \sin\alpha.$$

At the entry, the direction of motion turns towards the centre by an angle $\alpha - \beta$. Inside the sphere, the particle moves along a chord, which makes the same angle β with the radii at both of its ends. By the symmetry of the path with respect to the radius bisecting this chord, the particle leaves the sphere at the angle α with the normal and is again deflected by $\alpha - \beta$ in the same sense. The total angle of deflection is thus

$$\chi = 2(\alpha - \beta).$$

To express ρ in terms of χ, let $q = \chi/2$, so that $\beta = \alpha - q$. The law of refraction becomes

$$\sin(\alpha - q) = \sin\alpha\cos q - \cos\alpha \sin q = \frac{\sin\alpha}{n}.$$

Dividing by $\sin\alpha$,

$$\cos q - \cot\alpha \sin q = \frac{1}{n} \Rightarrow \cot\alpha = \frac{n\cos q - 1}{n \sin q}.$$

Using $\sin^2\alpha = 1/(1 + \cot^2\alpha)$, we get

$$\sin^2\alpha = \frac{n^2 \sin^2 q}{n^2 \sin^2 q + (n\cos q - 1)^2} = \frac{n^2 \sin^2 q}{n^2 + 1 - 2n\cos q},$$

since $n^2 \sin^2 q + n^2 \cos^2 q - 2n\cos q + 1 = n^2 + 1 - 2n\cos q$. Therefore

$$\rho^2 = a^2 \sin^2\alpha = \frac{a^2 n^2 \sin^2(\chi/2)}{n^2 + 1 - 2n\cos(\chi/2)}.$$

Let $D = n^2 + 1 - 2n\cos q$, so that $\mathrm{d}D/\mathrm{d}q = 2n\sin q$. Since $\mathrm{d}q/\mathrm{d}\chi = 1/2$, the quotient rule gives

$$\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi} = \frac{a^2n^2}{2}\frac{2\sin q\cos q\, D - \sin^2 q\cdot 2n\sin q}{D^2} = \frac{a^2n^2\sin q\left[\cos q\, D - n\sin^2 q\right]}{D^2}.$$

The bracket factorizes as

$$\begin{aligned}\cos q\, D - n\sin^2 q &= n^2\cos q + \cos q - 2n\cos^2 q - n + n\cos^2 q\\ &= n^2\cos q - n\cos^2 q - n + \cos q\\ &= (n\cos q - 1)(n - \cos q).\end{aligned}$$

We now check that χ increases with ρ. As ρ increases from 0 to a, α increases from 0 to $\pi/2$, and since $\sin\beta = \sin\alpha/n$,

$$\frac{\mathrm{d}\chi}{\mathrm{d}\alpha} = 2\left(1 - \frac{\mathrm{d}\beta}{\mathrm{d}\alpha}\right) = 2\left(1 - \frac{\cos\alpha}{n\cos\beta}\right) = 2\left(1 - \frac{\cos\alpha}{\sqrt{n^2 - \sin^2\alpha}}\right) > 0,$$

because $n^2 - \sin^2\alpha > 1 - \sin^2\alpha = \cos^2\alpha$. The relation between ρ and χ is thus one-to-one, and $\mathrm{d}(\rho^2)/\mathrm{d}\chi > 0$. The particles deflected into the range $\mathrm{d}\chi$ pass through an annulus of area

$$\mathrm{d}\sigma = \pi\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\,\mathrm{d}\chi = \frac{1}{2\sin\chi}\frac{\mathrm{d}(\rho^2)}{\mathrm{d}\chi}\,\mathrm{d}o,$$

where we used $\mathrm{d}o = 2\pi\sin\chi\,\mathrm{d}\chi$. With $\sin\chi = 2\sin q\cos q$, the factor $\sin q$ cancels, and

$$\mathrm{d}\sigma = \frac{a^2n^2(n\cos q - 1)(n - \cos q)}{4\cos q\, D^2}\,\mathrm{d}o.$$

Let us find the range of χ. For $\rho = 0$, the particle crosses the sphere along a diameter, so $\alpha = \beta = 0$ and $\chi = 0$. As $\rho \to a$ from below, the particle grazes the sphere, with

$$\alpha \to \frac{\pi}{2}, \qquad \beta \to \arcsin\frac{1}{n}.$$

Hence

$$\chi_{max} = 2\left(\frac{\pi}{2} - \arcsin\frac{1}{n}\right) = 2\arccos\frac{1}{n}, \qquad \cos\frac{\chi_{max}}{2} = \frac{1}{n}.$$

The formula for $\mathrm{d}\sigma$ applies for $0 < \chi < \chi_{max}$. Accordingly, the factor $n\cos(\chi/2) - 1$ is positive in this range and vanishes at $\chi = \chi_{max}$. Particles with $\rho > a$ are not deflected and do not contribute to the scattering.

Finally, the total cross-section for scattering is obtained directly from the area of the annuli,

$$\sigma = \int_0^{\chi_{max}} \mathrm{d}\sigma = \pi\left[\rho^2(\chi_{max}) - \rho^2(0)\right].$$

At $\chi = \chi_{max}$, $\cos q = 1/n$, so $\sin^2 q = (n^2 - 1)/n^2$ and $D = n^2 - 1$, which gives $\rho^2(\chi_{max}) = a^2$. Since $\rho^2(0) = 0$, $\sigma = \pi a^2$. This is simply the cross-sectional area of the sphere: every particle which hits the sphere is deflected. We thus get

Answer to Problem 13.

$$\mathrm{d}\sigma = \frac{a^2n^2\left[n\cos(\chi/2) - 1\right]\left[n - \cos(\chi/2)\right]}{4\cos(\chi/2)\left[n^2 + 1 - 2n\cos(\chi/2)\right]^2}\,\mathrm{d}o, \qquad n = \sqrt{1 + \frac{2U_0}{mv_\infty^2}},$$

$$0 < \chi < \chi_{max} = 2\arccos\frac{1}{n}, \qquad \sigma = \pi a^2$$

Problem 14.

Derive formula (20.3) from (18.4).

Solution: Consider the relative motion, with reduced mass m and relative velocity at infinity v_∞. Formula (18.4) of the book gives

$$\phi_0 = \int_{r_{min}}^{\infty} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}}, \qquad \chi = \pi - 2\phi_0.$$

For a weak field, it is tempting to expand the integrand in powers of U. However, the integrand behaves as $1/\sqrt{r-r_{min}}$ near the turning point. Expanding it gives terms like $U/(1-\rho^2/r^2)^{3/2}$, which are not integrable at $r=\rho$. These divergences are spurious: they only come from expanding near the point where the radicand vanishes.

To avoid this, we write ϕ_0 as the derivative of an integral whose integrand vanishes at the turning point. Since this integral does not converge at infinity, we first introduce a finite upper limit R,

$$J_R(\rho) = \int_{r_{min}}^{R} \sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}\,\mathrm{d}r.$$

We differentiate with respect to ρ at fixed R. The lower limit r_{min} depends on ρ, but its contribution is the integrand at $r=r_{min}$ times $\mathrm{d}r_{min}/\mathrm{d}\rho$, which vanishes since the square root is zero there. Hence

$$\frac{\partial J_R}{\partial \rho} = -\int_{r_{min}}^{R} \frac{\rho\,\mathrm{d}r}{r^2\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U(r)}{mv_\infty^2}}} \Rightarrow \phi_{0,R} = -\frac{\partial J_R}{\partial \rho},$$

where $\phi_{0,R}$ is the integral (18.4) cut off at $r=R$. We keep R finite until after the differentiation, since J_R itself grows like R and has no limit as $R\to\infty$.

We now expand J_R to first order in U. Pointwise,

$$\sqrt{1-\frac{\rho^2}{r^2}-\frac{2U}{mv_\infty^2}} \simeq \sqrt{1-\frac{\rho^2}{r^2}} - \frac{U}{mv_\infty^2\sqrt{1-\rho^2/r^2}} = \sqrt{1-\frac{\rho^2}{r^2}} - \frac{1}{mv_\infty^2}\frac{U(r)\,r}{\sqrt{r^2-\rho^2}}.$$

This expansion is not uniform near the turning point, but it can be integrated. To see why, regard J_R as a function of the strength of the field. Its first-order variation contains a term from the shift of the lower limit, from ρ (the turning point when $U=0$) to r_{min}. This term is proportional to the unperturbed integrand at its lower endpoint, which vanishes. The other term is the integral of the first-order variation of the integrand, which has only an integrable singularity $1/\sqrt{r-\rho}$. To first order, we can therefore take the lower limit at ρ,

$$J_R \simeq \int_\rho^R \sqrt{1-\frac{\rho^2}{r^2}}\,\mathrm{d}r - \frac{1}{mv_\infty^2}\int_\rho^R \frac{U(r)\,r\,\mathrm{d}r}{\sqrt{r^2-\rho^2}} = \int_\rho^R \sqrt{1-\frac{\rho^2}{r^2}}\,\mathrm{d}r - \frac{K_R(\rho)}{mv_\infty^2},$$

where

$$K_R(\rho) = \int_\rho^R \frac{U(r)\,r\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}.$$

The free term can be integrated directly,

$$\int_\rho^R \sqrt{1-\frac{\rho^2}{r^2}}\,\mathrm{d}r = \int_\rho^R \frac{\sqrt{r^2-\rho^2}}{r}\,\mathrm{d}r = \Big[\sqrt{r^2-\rho^2} - \rho\arccos\frac{\rho}{r}\Big]_\rho^R = \sqrt{R^2-\rho^2} - \rho\arccos\frac{\rho}{R},$$

as can be checked by differentiating the bracket with respect to r,

$$\frac{r}{\sqrt{r^2-\rho^2}} - \frac{\rho^2}{r\sqrt{r^2-\rho^2}} = \frac{\sqrt{r^2-\rho^2}}{r}.$$

Its derivative with respect to ρ is

$$\frac{\partial}{\partial\rho}\Big(\sqrt{R^2-\rho^2} - \rho\arccos\frac{\rho}{R}\Big) = -\frac{\rho}{\sqrt{R^2-\rho^2}} - \arccos\frac{\rho}{R} + \frac{\rho}{\sqrt{R^2-\rho^2}} = -\arccos\frac{\rho}{R}.$$

Consequently,

$$\phi_{0,R} \simeq \arccos\frac{\rho}{R} + \frac{1}{mv_\infty^2}\frac{\partial K_R}{\partial \rho}.$$

We cannot differentiate K_R directly under the integral sign, because its integrand is singular at the lower limit $r = \rho$. We therefore integrate by parts first, using

$$\frac{r\,\mathrm{d}r}{\sqrt{r^2-\rho^2}} = \mathrm{d}\sqrt{r^2-\rho^2},$$

which gives

$$K_R = \Big[U(r)\sqrt{r^2-\rho^2}\Big]_\rho^R - \int_\rho^R U'(r)\sqrt{r^2-\rho^2}\,\mathrm{d}r = U(R)\sqrt{R^2-\rho^2} - \int_\rho^R U'(r)\sqrt{r^2-\rho^2}\,\mathrm{d}r.$$

The boundary term at $r = \rho$ vanishes since $\sqrt{r^2-\rho^2} = 0$ there. The new integrand is finite at $r = \rho$, and it vanishes there. Hence, when we differentiate with respect to ρ, the variation of the lower limit again gives no contribution, and

$$\frac{\partial K_R}{\partial \rho} = -\frac{\rho U(R)}{\sqrt{R^2-\rho^2}} + \rho\int_\rho^R \frac{U'(r)\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}.$$

Now we take $R \to \infty$. Then $\arccos(\rho/R) \to \pi/2$, and the first term vanishes since $U(R) \to 0$ and $\rho/\sqrt{R^2-\rho^2} \to 0$. Assuming the remaining integral converges (*i.e.* the force decreases fast enough), we get

$$\phi_0 \simeq \frac{\pi}{2} + \frac{\rho}{mv_\infty^2}\int_\rho^\infty \frac{U'(r)\,\mathrm{d}r}{\sqrt{r^2-\rho^2}},$$

and therefore

$$\chi = \pi - 2\phi_0 \simeq -\frac{2\rho}{mv_\infty^2}\int_\rho^\infty \frac{U'(r)\,\mathrm{d}r}{\sqrt{r^2-\rho^2}}.$$

For a repulsive field, $U' < 0$ and $\chi > 0$, as it should be. This result is valid only if the deflection is small, *i.e.* if the field is weak along the whole path ($|U| \ll \frac{1}{2}mv_\infty^2$ for $r \gtrsim \rho$). Under this condition, the first-order expansion is justified.

The angle χ is the scattering angle in the centre-of-mass system. Formula (20.3) gives instead θ_1, the deflection of the incident particle in the laboratory system, where the particle m_2 is initially at rest. From equation (17.4) of the book, for small angles

$$\tan\theta_1 = \frac{m_2\sin\chi}{m_1+m_2\cos\chi} \Rightarrow \theta_1 \simeq \frac{m_2}{m_1+m_2}\chi = \frac{m}{m_1}\chi, \qquad m = \frac{m_1m_2}{m_1+m_2}.$$

Multiplying χ by m/m_1, the reduced mass cancels. We thus get

Answer to Problem 14.

$$\theta_1 \simeq -\frac{2\rho}{m_1v_\infty^2}\int_\rho^\infty \frac{\mathrm{d}U}{\mathrm{d}r}\frac{\mathrm{d}r}{\sqrt{r^2-\rho^2}},$$
$$\chi \simeq -\frac{2\rho}{mv_\infty^2}\int_\rho^\infty \frac{\mathrm{d}U}{\mathrm{d}r}\frac{\mathrm{d}r}{\sqrt{r^2-\rho^2}}$$

Problem 15.

Determine the effective cross-section for small-angle scattering in a field $U = \alpha/r^n$ $(n > 0)$.

Solution: We take $\alpha > 0$ (a repulsive field). Let m_2 be the mass of the scattering particle, initially at rest, and

$$m = \frac{m_1 m_2}{m_1 + m_2}$$

the reduced mass. As shown in Problem 14, when the scattering angle χ in the centre-of-mass system is small ($\chi \ll 1$), the deflection of the incident particle of mass m_1 is $\theta_1 \simeq (m/m_1)\chi$, and it is given by formula (20.3) of the book,

$$\theta_1 \simeq -\frac{2\rho}{m_1 v_\infty^2} \int_\rho^\infty \frac{\mathrm{d}U}{\mathrm{d}r} \frac{\mathrm{d}r}{\sqrt{r^2 - \rho^2}}.$$

Since

$$\frac{\mathrm{d}U}{\mathrm{d}r} = -\frac{n\alpha}{r^{n+1}},$$

the deflection is

$$\theta_1 \simeq \frac{2n\alpha\rho}{m_1 v_\infty^2} \int_\rho^\infty \frac{\mathrm{d}r}{r^{n+1}\sqrt{r^2 - \rho^2}}.$$

To evaluate the integral, we substitute

$$u = \frac{\rho^2}{r^2}, \qquad r = \rho u^{-1/2}, \qquad \mathrm{d}r = -\frac{\rho}{2} u^{-3/2}\,\mathrm{d}u.$$

When r goes from ρ to ∞, u goes from 1 to 0, and the minus sign of $\mathrm{d}r$ is absorbed by reversing the limits. We also have

$$r^{n+1} = \rho^{n+1} u^{-(n+1)/2}, \qquad \sqrt{r^2 - \rho^2} = \rho\sqrt{\frac{1}{u} - 1} = \rho u^{-1/2}(1-u)^{1/2}.$$

Collecting the powers of u, $-\frac{3}{2} + \frac{n+1}{2} + \frac{1}{2} = \frac{n-1}{2}$, so

$$\int_\rho^\infty \frac{\mathrm{d}r}{r^{n+1}\sqrt{r^2 - \rho^2}} = \frac{1}{2\rho^{n+1}} \int_0^1 u^{(n-1)/2}(1-u)^{-1/2}\,\mathrm{d}u.$$

This is the Beta function $B(p,q) = \int_0^1 u^{p-1}(1-u)^{q-1}\,\mathrm{d}u = \Gamma(p)\Gamma(q)/\Gamma(p+q)$ with $p = (n+1)/2$ and $q = 1/2$. Since $\Gamma(1/2) = \sqrt{\pi}$,

$$\int_\rho^\infty \frac{\mathrm{d}r}{r^{n+1}\sqrt{r^2 - \rho^2}} = \frac{1}{2\rho^{n+1}} B\left(\frac{n+1}{2}, \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2\rho^{n+1}} \frac{\Gamma\left((n+1)/2\right)}{\Gamma\left((n+2)/2\right)}.$$

Using $\Gamma\left((n+2)/2\right) = \frac{n}{2}\Gamma(n/2)$, the factor n cancels and

$$\theta_1 \simeq \frac{2\sqrt{\pi}\,\alpha}{m_1 v_\infty^2 \rho^n} \frac{\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}.$$

Let us define

$$C_n = \frac{2\sqrt{\pi}\,\alpha}{m_1 v_\infty^2} \frac{\Gamma\left((n+1)/2\right)}{\Gamma(n/2)},$$

so that $\theta_1 \simeq C_n/\rho^n$. The deflection decreases monotonically with ρ, and the relation can be inverted,

$$\rho \simeq \left(\frac{C_n}{\theta_1}\right)^{1/n}, \qquad \left|\frac{\mathrm{d}\rho}{\mathrm{d}\theta_1}\right| = \frac{1}{n} C_n^{1/n} \theta_1^{-1/n-1} = \frac{\rho}{n\theta_1}.$$

The particles deflected into the range $\mathrm{d}\theta_1$ pass through an annulus of area $2\pi\rho|\mathrm{d}\rho|$ in the incident beam. For small angles, the element of solid angle is

$$\mathrm{d}o_1 = 2\pi \sin\theta_1\,\mathrm{d}\theta_1 \simeq 2\pi\theta_1\,\mathrm{d}\theta_1,$$

so

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o_1} \simeq \frac{\rho}{\theta_1}\left|\frac{\mathrm{d}\rho}{\mathrm{d}\theta_1}\right| = \frac{\rho^2}{n\theta_1^2} = \frac{1}{n}\frac{C_n^{2/n}}{\theta_1^{2+2/n}}.$$

This result requires weak scattering in the centre-of-mass system, $\chi \simeq (m_1/m)\theta_1 \ll 1$. Multiplying $\theta_1 \simeq C_n/\rho^n$ by m_1/m, the corresponding centre-of-mass deflection is

$$\chi \simeq \frac{2\sqrt{\pi}\,\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}\frac{\alpha/\rho^n}{mv_\infty^2},$$

whose numerical coefficient is of order unity for n of order unity. The condition $\chi \ll 1$ is thus the weak-field condition

$$\frac{\alpha}{\rho^n} \ll \frac{1}{2}mv_\infty^2,$$

i.e. the potential energy at the distance of closest approach $r \simeq \rho$ must be small compared with the energy of the relative motion. Since $m \leq m_1$, it implies $\theta_1 \ll 1$, which justifies $\mathrm{d}o_1 \simeq 2\pi\theta_1\,\mathrm{d}\theta_1$. However, the converse does not hold for arbitrary masses. For instance, if $m_1 > m_2$, the laboratory angle never exceeds $\arcsin(m_2/m_1)$, so it can be small even when χ is not. The two conditions coincide only when $m_2 \gg m_1$, where $m \simeq m_1$ and $\theta_1 \simeq \chi$. The result must not be extended to arbitrary angles. Note also that $\mathrm{d}\sigma$ diverges as $\theta_1 \to 0$, so the total cross-section is infinite unless the angles are cut off.

As a check, for $n = 1$ we have $\Gamma(1)/\Gamma(1/2) = 1/\sqrt{\pi}$, so $C_1 = 2\alpha/(m_1 v_\infty^2)$ and

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o_1} \simeq \frac{4\alpha^2}{m_1^2 v_\infty^4 \theta_1^4},$$

which is the small-angle limit of the Rutherford formula. For $n = 2$, $\Gamma(3/2)/\Gamma(1) = \sqrt{\pi}/2$, so $C_2 = \pi\alpha/(m_1 v_\infty^2)$ and

$$\frac{\mathrm{d}\sigma}{\mathrm{d}o_1} \simeq \frac{\pi\alpha}{2m_1 v_\infty^2 \theta_1^3},$$

in agreement with the small-angle limit found in Problem 12 for a fixed centre ($m_1 = m$, $\theta_1 = \chi$). We thus get

Answer to Problem 15.

$$\theta_1 \simeq \frac{2\sqrt{\pi}\,\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}\frac{\alpha}{m_1 v_\infty^2 \rho^n},$$

$$\mathrm{d}\sigma \simeq \frac{1}{n}\left[\frac{2\sqrt{\pi}\,\Gamma\left((n+1)/2\right)}{\Gamma(n/2)}\frac{\alpha}{m_1 v_\infty^2}\right]^{2/n}\frac{\mathrm{d}o_1}{\theta_1^{2+2/n}},$$

$$\text{valid for } \frac{\alpha}{\rho^n} \ll \frac{1}{2}mv_\infty^2, \qquad m = \frac{m_1 m_2}{m_1 + m_2}$$

Chapter 5.

Small Oscillations

Chapter 6.

Motion of a Rigid Body

Problem 1.

Determine the principal moments of inertia for the following types of molecule, regarded as systems of particles at fixed distances apart:

(a)

a molecule of collinear atoms

Solution: Choose the x_3-axis along the line containing all atoms, and place the origin at the centre of mass. Then each particle has coordinates $(0, 0, x_a)$, so the inertia tensor is diagonal in the basis (x_1, x_2, x_3), with

$$I_3 = 0, \qquad I_1 = I_2.$$

Indeed, since the distance to the x_1-axis (or x_2-axis) is $|x_a - x_{CM}|$ for each particle,

$$I_1 = I_2 = \sum_a m_a \, (x_a - x_{CM})^2, \qquad x_{CM} = \frac{1}{\mu} \sum_a m_a x_a, \quad \mu = \sum_a m_a.$$

Expanding and using the definition of x_{CM} gives

$$\begin{aligned} I_1 = I_2 &= \sum_a m_a x_a^2 - 2 x_{CM} \sum_a m_a x_a + \mu x_{CM}^2 \\ &= \sum_a m_a x_a^2 - \frac{1}{\mu} \left(\sum_a m_a x_a \right)^2 . \end{aligned}$$

To rewrite this in terms of pairwise separations, expand the square:

$$\left(\sum_a m_a x_a \right)^2 = \sum_a m_a^2 x_a^2 + 2 \sum_{a<b} m_a m_b x_a x_b .$$

Hence

$$\begin{aligned} I_1 = I_2 &= \sum_a m_a x_a^2 - \frac{1}{\mu} \sum_a m_a^2 x_a^2 - \frac{2}{\mu} \sum_{a<b} m_a m_b x_a x_b \\ &= \frac{1}{\mu} \sum_a m_a (\mu - m_a) x_a^2 - \frac{2}{\mu} \sum_{a<b} m_a m_b x_a x_b . \end{aligned}$$

Now note that $\mu - m_a = \sum_{b \neq a} m_b$, so

$$\sum_a m_a (\mu - m_a) x_a^2 = \sum_a \sum_{b \neq a} m_a m_b x_a^2 = \sum_{a<b} m_a m_b (x_a^2 + x_b^2).$$

Therefore

$$\begin{aligned} I_1 = I_2 &= \frac{1}{\mu} \sum_{a<b} m_a m_b (x_a^2 + x_b^2) - \frac{2}{\mu} \sum_{a<b} m_a m_b x_a x_b \\ &= \frac{1}{\mu} \sum_{a<b} m_a m_b (x_a - x_b)^2 . \end{aligned}$$

Since for collinear atoms the pair distance is $l_{ab} = |x_a - x_b|$, we finally obtain

Answer to Problem 1 (a).

$$I_1 = I_2 = \frac{1}{\mu} \sum_{a<b} m_a m_b \, l_{ab}^2$$

$$I_3 = 0$$

(b)

a triatomic molecule which is an isosceles triangle (Fig. 36)

Solution: Let the molecule lie in the (x_1, x_2)-plane, with x_1 along the base and x_2 along the axis of symmetry. The two base atoms have equal masses m_1 and are separated by a distance a; the apex atom has mass m_2 and is at height h above the base midpoint. Denote the total mass by

$$\mu = 2m_1 + m_2.$$

Choose coordinates (before shifting to the centre of mass):

$$(x_1, x_2)_{(1)} = \left(-\frac{a}{2}, 0\right), \qquad (x_1, x_2)_{(3)} = \left(+\frac{a}{2}, 0\right), \qquad (x_1, x_2)_{(2)} = (0, h),$$

where atoms (1) and (3) have mass m_1 and atom (2) has mass m_2. By symmetry $x_{1,CM} = 0$, and

$$x_{2,CM} = \frac{m_2 h}{\mu}.$$

The symmetry implies that the axes through the centre of mass parallel to x_1, x_2, and x_3 are principal. Hence,

$$I_1 = \sum_a m_a (x_{2,a} - x_{2,CM})^2, \qquad I_2 = \sum_a m_a (x_{1,a} - x_{1,CM})^2, \qquad I_3 = I_1 + I_2,$$

since the molecule is planar ($x_3 \equiv 0$). Using the pair-distance identity in one dimension,

$$\sum_a m_a (x_{2,a} - x_{2,CM})^2 = \frac{1}{\mu} \sum_{a<b} m_a m_b (x_{2,a} - x_{2,b})^2,$$

and noting that the only non-zero vertical separations are between the apex and each base atom, equal to h, we obtain

$$I_1 = \frac{1}{\mu}\left(m_1 m_2 h^2 + m_1 m_2 h^2\right) = \frac{2m_1 m_2}{\mu}\, h^2.$$

Similarly,

$$I_2 = \sum_a m_a (x_{1,a} - x_{1,CM})^2 = \frac{1}{\mu} \sum_{a<b} m_a m_b (x_{1,a} - x_{1,b})^2.$$

The horizontal separations are between the two base atoms, and between the apex and either base atom. Hence

$$I_2 = \frac{1}{\mu}\left(m_1^2 a^2 + 2m_1 m_2 \left(\frac{a}{2}\right)^2\right) = \frac{1}{\mu}\left(m_1^2 a^2 + \frac{1}{2} m_1 m_2 a^2\right) = \frac{1}{2} m_1 a^2.$$

Therefore

$$I_3 = I_1 + I_2 = \frac{2m_1 m_2}{\mu}\, h^2 + \frac{1}{2} m_1 a^2.$$

Answer to Problem 1 (b).

$$I_1 = \frac{2m_1 m_2}{\mu}\, h^2$$
$$I_2 = \frac{1}{2} m_1 a^2$$
$$I_3 = I_1 + I_2.$$

(c)

a tetratomic molecule which is an equilateral-based tetrahedron

Solution: The base is an equilateral triangle of side a in the plane $x_3 = 0$, with three equal masses m_1 at its vertices. A fourth atom of mass m_2 lies on the symmetry axis through the base centroid at height h above the base. Let

$$\mu = 3m_1 + m_2.$$

Let ρ be the distance from the base centroid to a base vertex. For an equilateral triangle,

$$\rho = \frac{a}{\sqrt{3}}, \qquad \rho^2 = \frac{a^2}{3}.$$

By symmetry the centre of mass lies on the symmetry axis. Measuring x_3 from the base plane,

$$x_{3,CM} = \frac{m_2 h}{\mu}.$$

The symmetry axis is a principal axis; any two orthogonal axes through the centre of mass perpendicular to it are also principal and have equal moments. Thus

$$I_1 = I_2, \qquad I_3 \text{ about the symmetry axis } (x_3).$$

The apex mass lies on the axis and contributes nothing. Each base mass is at distance ρ from the axis, hence

$$I_3 = 3m_1\rho^2 = 3m_1\frac{a^2}{3} = m_1 a^2.$$

Let x_1 be any axis through the centre of mass perpendicular to x_3. Then

$$I_1 = \sum_a m_a\,(y_a^2 + z_a^2),$$

where y is a coordinate orthogonal to x_1 in the plane perpendicular to the axis, and z is measured along x_3 from the centre of mass. The vertical coordinates relative to the centre of mass are

$$z_{\text{base}} = -x_{3,CM} = -\frac{m_2 h}{\mu}, \qquad z_{\text{apex}} = h - x_{3,CM} = h\left(1 - \frac{m_2}{\mu}\right) = \frac{3m_1}{\mu}h.$$

For the three vertices of an equilateral triangle, for any in-plane axis through the centroid,

$$\sum_{\text{base}} y^2 = \frac{3}{2}\rho^2 = \frac{3}{2}\cdot\frac{a^2}{3} = \frac{1}{2}a^2.$$

Therefore,

$$\begin{aligned} I_1 &= m_1\left(\sum_{\text{base}} y^2\right) + 3m_1 z_{\text{base}}^2 + m_2 z_{\text{apex}}^2 \\ &= \frac{1}{2}m_1 a^2 + 3m_1\left(\frac{m_2^2 h^2}{\mu^2}\right) + m_2\left(\frac{9m_1^2 h^2}{\mu^2}\right) \\ &= \frac{1}{2}m_1 a^2 + \frac{3m_1 m_2}{\mu}\,h^2. \end{aligned}$$

Hence $I_2 = I_1$.

Answer to Problem 1 (c).

$$I_1 = I_2 = \frac{3m_1 m_2}{\mu}\,h^2 + \frac{1}{2}m_1 a^2$$
$$I_3 = m_1 a^2$$